Data Driven Approach for Measuring Child Development

A Dual Degree Report

Submitted in fulfilment of the requirements

for

Interdisciplinary Dual Degree (B.Tech + M.Tech)

in

Centre for Machine Intelligence and Data Science

by

Abhishek Choudhary

Roll No.: 210010003

Under the guidance of

Prof. Sharat Chandran

Indian Institute of Technology Bombay

Abstract

Early childhood is a period of rapid cognitive, motor, and social development, and reliable measurement of these developmental trajectories is essential for identifying children at risk and informing early intervention. Existing assessment tools combine rich item-level response data with expert-designed scoring rubrics. While these rubrics are often clinically grounded, they may not fully exploit the structure latent in the response data; further they require adaptation across cultural and developmental contexts.This report develops a data-driven framework for measuring child development directly from item-level responses from approximately 4,000 subjects aged 0-6, with the vast majority of children exhibiting prototypically neurotypical development.

We propose an unsupervised probabilistic generative model that combines traditional psychometric statistical methods with a Variational Autoencoder (VAE). The encoder maps high-dimensional response patterns to a low-dimensional latent ability space, and the decoder is constrained to follow a 2-Parameter Logistic model for binary items and a graded response model for ordinal items. Training is fast and runs on commodity hardware.We validate the learned latent measures along three axes. First, they are strongly monotonic in chronological age, confirming that they track genuine developmental progress. Second, they correlate in the expected directions with adverse-childhood and protective covariates. Third, a thin neural network maps latent representations to the Griffiths Mental Development Scales (GMDS), establishing criterion validity against an established clinical instrument. Finally, we then propose a deployment strategy: MDAT for ages 0–3 and DEEP for ages 3–6, motivated by training cost, age-appropriateness, and alignment with the WHO Global Scales for Early Development (GSED).

Acknowledgement

I express my heartfelt gratitude to **Prof. Sharat Chandran** for his guidance and support throughout the course of this project. I would also like to thank **Kumar Ankur** and **Praveen Maurya** for their support. I am also sincerely grateful to **Prof. Bhismadev Chakrabarti** (University of Reading), **Dr. Supriya Bhavnani** (Sangath) and **Dr. Gareth McCray** (University of Keele) for their valuable collaboration, insights, and support during the course of this research. Additionally, I am grateful to the CMInDS department for providing high-quality education and infrastructure for research pursuits. I would also like to thank my friends and peers at IIT Bombay for their encouragement and support throughout my academic journey. Finally, I am deeply grateful to my family for their constant motivation, patience, and unwavering support.

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

I declare that I have used the following AI tools and/or other similar tools:

ChatGPT (OpenAI), Claude (Anthropic), and GitHub Copilot

for the purpose of:

- Assistance with coding tasks, including debugging, refactoring, and generating boilerplate for the Variational Autoencoder, psychometric models (2PL and graded response models), and data preprocessing pipelines used in this work.
- Theoretical study and clarification of concepts in psychometrics (Item Response Theory, graded response models), variational inference, and deep generative models, used as a supplementary learning aid alongside primary literature.
- Improving the clarity, grammar, and language of selected portions of this report and creating some of the images used. All technical content, experimental design, results, and conclusions are my own.

Abhishek Choudhary

Contents

Abstract **i**

Acknowledgement **ii**

Declaration **iii**

1 Introduction **1**

1.1 Motivation 1
1.2 Challenge 1
1.3 Solution Approach 2
1.4 Summary of Contributions 3
1.5 Report Organisation 4

2 Literature Survey **6**

2.1 Developmental Assessment Instruments 6
2.2 Item Response Theory 7
2.3 Variational Autoencoders 7
2.4 Hybrid IRT–Deep Learning Approaches 7
2.5 Position of This report 8

3 Background: The IRT–VAE Framework **9**

3.1 Why Combine IRT and VAEs 9
3.2 Encoder: Approximating the Posterior 10
3.3 Decoder: An IRT Likelihood, Not a Neural Net 10
3.4 Training Objective 10
3.5 Ordinal Items: The Graded Response Model 11
3.6 Handling Missing Responses 11

4 Datasets **13**

4.1 The STREAM Initiative 13
4.2 MDAT: Malawi Developmental Assessment Tool 14
4.2.1 Deriving Latent Scores from MDAT 15
4.3 DEEP: Developmental Assessment on an E-Platform 15

4.3.1 Deriving Latent Scores from DEEP 16
4.4 START: Screening Tool for Autism Risk using Technology 17
4.5 Convergent Variables 18
4.6 Criterion Variable: GMDS 19

5 Validation of Latent Representations 21
5.1 Do Latent Measures Track Chronological Age? 21
5.2 Are Latent Measures Sensitive to Adverse Childhood Experiences? 22
5.2.1 Why Age-Adjustment Is Necessary 22
5.2.2 Convergent Validity Results 23
5.3 Is Chronological Age Itself a Sufficient Measure? 24

6 Deployment Strategy and Validity 26
6.1 An Age-Split Deployment: MDAT for 0–3, DEEP for 3–6 26
6.2 Criterion Validity via GMDS 27
6.3 Convergent Validity of Predicted GMDS Scores 28
6.3.1 Mapping MDAT to GMDS, Ages 0–3 28
6.3.2 Mapping DEEP to GMDS, Ages 3–6 29
6.4 Convergent Validity of GMDS Itself 30
6.5 Convergent Validity of Age-Split Latent Totals 31
6.5.1 MDAT Total, Ages 0–3 32
6.5.2 DEEP Total, Ages 3–6 32
6.6 Summary of the Deployment Strategy 33

7 Operational Refinements: Reducing DEEP and START for Screening 34
7.1 Reducing DEEP Administration Time via Game Subset Selection 34
7.1.1 The Problem and the Combinatorial Wall 34
7.1.2 Decomposition: Singleton Latents Plus a Combining Network . . 35
7.1.3 Technical Merits of the Decomposition 35
7.1.4 Results: Five Games Match Fourteen 36
7.1.5 Open Questions 36
7.2 START: Continuous IRT–VAE and the Age Test 37
7.2.1 Continuous IRT–VAE for START 37
7.2.2 The Age Test 38
7.2.3 Does START Add Anything to MDAT + DEEP for GMDS? . . . 39
7.2.4 Conclusion 40
7.3 START as a Screening Tool 41
7.3.1 Setup: Class Imbalance and SMOTE 41
7.3.2 START Alone: Aggregate Performance 41
7.3.3 Solo Game Performance 42

7.3.4 Augmenting MDAT + DEEP with START Games 42
7.3.5 Implications for Deployment 44
7.3.6 Conclusion: A Selective Screener 44

8 Age-Stratified Modality Selection 45
8.1 Single- and Combined-Modality Performance (Toddlers, 2.5–6) 45
8.2 The 4–6 Band: Where MDAT Loses Ground 46
8.3 Infants and Younger Toddlers (0–6 and 0–2.5) 47
8.4 Summary 48

9 Robustness: Proxying and Generalisation 49
9.1 DEEP as a Proxy for MDAT 49
9.2 Cross-Country Generalisation: India vs. Malawi 50
9.2.1 Is the Difference Significant? 50
9.2.2 What Drives the Gap? 51
9.3 Summary 51

10 Conclusion 52
10.1 Summary 52
10.2 Limitations 53
10.3 Future Work 53

A VAE Mathematical Derivations 55
A.1 The Intractable Posterior 55
A.2 Encoder: Variational Approximation 55
A.3 Decoder 55
A.4 Derivation of the ELBO 56
A.5 Reparameterisation Trick 56
A.6 Monte Carlo Estimator 56
A.7 IRT–VAE Objective: 2PL 57
A.8 IRT–VAE Objective: GRM (Ordinal) 57
A.9 Masked Reconstruction 58

B GAMLSS: Background 59
B.1 Motivation 59
B.2 The GAMLSS Family 59
B.3 Use in This report 60

C Stage 1: Empirical Equivalence of the IRT–VAE with Classical IRT 61
C.1 MDAT 61
C.2 DEEP 62

C.3 Takeaway 62

List of Figures

1.1 Pictorial depiction of the work presented in this report 5

4.1 MDAT tasks example. The instrument is clinician-administered: each task is presented to the child in a standardised way and scored as success / failure on a paper protocol. 14
4.2 DEEP games. The child plays a sequence of 14 short games on a tablet. . 15
4.3 Representative tasks from the START battery. 17
4.4 Griffiths Mental Development Scales (GMDS). 19

5.1 Latent Total score against chronological age. Both MDAT and DEEP show the expected positive trend. MDAT grows steadily across the full 0–6 range; DEEP, which is an older-child instrument, shows a flatter early region and steeper growth in 3–6. 21

6.1 Responses are encoded and passed through a Neural Network to map to GMDS (Representative Picture) 27

7.1 START Total latent score (PLT only) against chronological age, ages 0–2.5. 38
7.2 START Total latent score (all the tasks) against chronological age, ages 2.5–6. 39

List of Tables

5.1 Correlation of Total latent scores with chronological age. 22
5.2 Convergent validity: correlation of age-adjusted Total latent scores with convergent variables. 23

6.1 Criterion validity: 5-fold CV correlation of predicted vs. actual GMDS raw scores. MDAT: $n = 462$; DEEP: $n = 484$. 28

6.2 Convergent validity of the predicted GMDS General score from MDAT (0–3 yrs). 29
6.3 Convergent validity of the predicted GMDS General score from DEEP (3–6 yrs). 29
6.4 Convergent validity of the GMDS General score (combined India + Malawi sample), 0–3 yrs. 30
6.5 Convergent validity of the GMDS General score (combined India + Malawi sample), 3–6 yrs. 31
6.6 Convergent validity: MDAT Total latent score, age-adjusted, ages 0–3. . 32
6.7 Convergent validity: DEEP Total latent score, age-adjusted, ages 3–6. . . 32

7.1 Criterion validity for predicting GMDS subscales from DEEP latents: full 14-game battery vs. 5-game subset. 36
7.2 Correlation of START Total (PLT only) with chronological age, ages 0–2.5. 39
7.3 Correlation of START Total (all the tasks) with chronological age, ages 2.5–6. 39
7.4 Predicting GMDS subscales using START alone (N = 949, 5-fold CV). . 40
7.5 Predicting GMDS subscales using MDAT + DEEP, no START ($n = 560$, 5-fold CV). 40
7.6 Predicting GMDS subscales using MDAT + DEEP + START ($n = 560$, 5-fold CV). 40
7.7 START-only screening baseline (16 features, no MDAT/DEEP). Best CV ROC-AUC in bold. 42
7.8 Solo START game screening performance (best classifier per game). Sens. = sensitivity; PPV = positive predictive value. 42
7.9 Anchor: MDAT + DEEP totals for screening (no START). Best CV ROC-AUC in bold. 43
7.10 Change in screening performance when one START game is added to MDAT + DEEP (best of RF/GB). Positive numbers indicate improvement over the anchor. 43
7.11 MDAT + DEEP + Wheel Task screening performance ($N = 2{,}525$, 67 enriched). 43

8.1 Single-modality criterion validity. 45
8.2 Combined-modality criterion validity. 46
8.3 Criterion validity in the 4–6 age band (5-fold CV). 46
8.4 Criterion validity for the 0–6 age band ($N \sim 950$). PLT is the Preferential Looking Task subset of START. 47
8.5 Criterion validity for the 0–2.5 age band ($N \sim 400$). PLT is the Preferential Looking Task subset of START. 47

9.1 Predicting MDAT scores from DEEP latent representations (5-fold CV). 49
9.2 DEEP criterion validity, India vs. Malawi (5-fold CV). 50
9.3 Fisher's z test of the India vs. Malawi difference, one-sided. 51

C.1 Correlation between Classical IRT (`mirt`) and IRT–VAE ability scores on MDAT. 61
C.2 Correlation between Classical IRT and IRT–VAE ability scores on DEEP. 62

Chapter 1

Introduction

1.1 Motivation

Early childhood—broadly the period from birth to six years of age—is the most plastic and consequential window in human development. It is during this window that the architecture of cognitive, motor, social, and linguistic abilities is laid down. Deficits accumulated during this period are difficult and expensive to remediate later in life. The corollary is that early, accurate identification of children whose development is faltering is among the highest-leverage interventions available in public health.

Globally, however, hundreds of millions of children under five live in environments—low income, high adversity, limited healthcare access—where their developmental potential is at risk. Measuring development at scale in such settings is therefore not a niche concern but a central one. Any framework that can produce reliable, fine-grained developmental scores from minimal input, on commodity hardware, and across cultural contexts, is of direct value to clinicians, public-health agencies, and policy planners.

1.2 Challenge

Despite the importance of the problem, measuring child development is intrinsically hard for several reasons.

Multidimensionality. Development is not a single quantity. A two-year-old may be ahead on language but behind on gross motor skills. A single composite score loses information that a clinician would use; a long battery of separate scores is hard to interpret in aggregate.

Reliance on trained assessors. The most rigorous instruments—Bayley Scales, Griffiths Mental Development Scales (GMDS), the Malawi Developmental Assessment Tool

(MDAT)—are administered face-to-face by trained clinicians. In low- and middle-income settings, the bottleneck is rarely the child or the test; it is the assessor.

Subjectivity and cultural calibration. Scoring rubrics are designed by experts based on theoretical models and normative samples. They embed cultural assumptions about what a "typical" three-year-old does. Items that travel poorly across cultures introduce noise.

Sparse and noisy data. Real assessment data is missing-not-at-random: the test stops when a child fails several items in a row, and starts at an age-dependent point. Naive scoring—summing items—confuses missingness with failure, and gives no principled way to share statistical strength across items.

The validation problem. Even if one builds a new measure, how does one know it is meaningful? Correlation with chronological age is a sanity check but not a sufficient one—age trivially correlates with everything that grows. Correlation with an established clinical instrument like GMDS is desirable, but GMDS itself is expensive, available only on a small subset, and noisy at the item level.

The challenge, then, is to derive developmental measures that are scalable, multidimensional, robust to missingness, statistically principled, and externally validated against both clinical instruments and known developmental determinants.

1.3 Solution Approach

We pose developmental measurement as the problem of learning, in an unsupervised manner, a low-dimensional latent representation $\mathbf{z} \in \mathbb{R}^k$ from a high-dimensional vector of observed item responses $\mathbf{x}$. The mapping $\mathbf{x} \mapsto \mathbf{z}$ is parameterised by a neural-network encoder; the reverse mapping—used as a likelihood model during training—is constrained to a 2PL IRT form for binary items and a GRM form for ordinal items. Missingness is handled by masking the reconstruction loss, so missing entries do not have to be imputed.

This single architectural choice yields several benefits in one step. The encoder is fast and amortised: at test time, scoring a new child is a single forward pass. The IRT-shaped decoder makes item parameters interpretable as difficulty and discrimination. The Gaussian prior in the latent space regularises the representation. And the entire model trains on a laptop in minutes for the dataset sizes we consider.

We then validate the learned latent scores in three ways:

(i) **Age monotonicity.** The Total latent score should increase with chronological age. We confirm this for both MDAT and DEEP.

(ii) **Convergent validity.** After removing the age trend with GAMLSS, the residual scores should correlate in the expected direction with established developmental determinants (HAZ, SES, FCI, MORS, CPRS, RBSK, CPAS). We confirm this.

(iii) **Criterion validity.** A small neural network mapping our latent scores to GMDS subscale scores should predict well in held-out cross-validation. We confirm this and use the predicted GMDS scores as a clinically-interpretable bridge.

Building on these validations, we propose a deployment strategy that uses MDAT for ages 0–3 and DEEP for ages 3–6, supported by criterion and convergent validity in both age-bands. We treat the third instrument, START, as a binary screening tool rather than a continuous scoring tool, because including START features in the regression to GMDS does not *enhances* performance. We characterise where START is most informative as a screener through per-task analyses.

Finally, we conduct a battery of cross-modal, age-stratified, and cross-country experiments that probe the limits of the framework: how well DEEP can stand in as a proxy for MDAT, where MDAT loses precision (later ages), and whether DEEP generalises across the Indian and Malawian cohorts.

1.4 Summary of Contributions

The contributions of this report are:

1. An IRT–VAE generative model for child-development data that handles binary and ordinal items, missingness, and scales to thousands of subjects on commodity hardware. A continuous-response variant uses a Gaussian IRF and is applied to START.

2. Validation of the learned latent scores against (a) chronological age, (b) age-adjusted convergent variables via GAMLSS, and (c) GMDS via a learned bridge model.

3. A practical deployment recipe (MDAT for 0–3, DEEP for 3–6) with quantified criterion and convergent validity in each age-band, and an analysis showing that wrong-signed cells in the predicted-GMDS convergent tables are inherited from GMDS itself on this sample, not introduced by the bridge.

4. A decomposed scoring pipeline—14 per-game IRT–VAEs feeding a shared combining network—that turns DEEP subset selection from a 2^{14}-VAE-retraining problem into a sweep of cheap NN regressions, and identifies a 5-game subset that matches the full 14-game battery on GMDS criterion validity, cutting DEEP administration time significantly.

5. Exclusion of START from the scoring pipeline, justified by two independent checks: the START Total latent correlates with age at only $r = 0.33$ for subjectes aged 0 to 2.5 years and $r = 0.52$ for the subjects with age 2.5 to 6 years, and conditional on MDAT and DEEP it adds nothing to GMDS prediction.

6. Cross-modal, age-stratified, and cross-country experiments that quantify how each instrument behaves alone, in combination, across age strata, and across the Indian and Malawian cohorts.

1.5 Report Organisation

The remainder of the report is organised as follows. Chapter 2 is a literature survey covering developmental assessment instruments, IRT, VAEs, and hybrid IRT–deep-learning models. Chapter 3 presents the IRT–VAE framework at the level of intuition needed to follow the rest of the report; the mathematical derivations are deferred to Chapter A. Chapter 4 describes the STREAM datasets (MDAT, DEEP, START) and the convergent and criterion variables (HAZ, SES, FCI, MORS, CPRS, CPAS, RBSK, GMDS). Chapter 5 validates the learned latent representations along the three axes above. Chapter 6 proposes and validates the age-split deployment strategy, including a check on GMDS's own convergent profile. Chapter 7 addresses the two operational gaps in the deployment: it shortens DEEP via game-subset selection, and excludes START using a continuous-response IRT–VAE and the age-correlation test. Chapter 8 and Chapter 9 reports the cross-modal, age-stratified, and cross-country experiments. Chapter 10 concludes and discusses future work.

Appendix A gives the full VAE derivations, Appendix B covers GAMLSS, and Appendix C reports the Stage 1 empirical equivalence of the IRT–VAE with classical IRT that motivated its adoption.

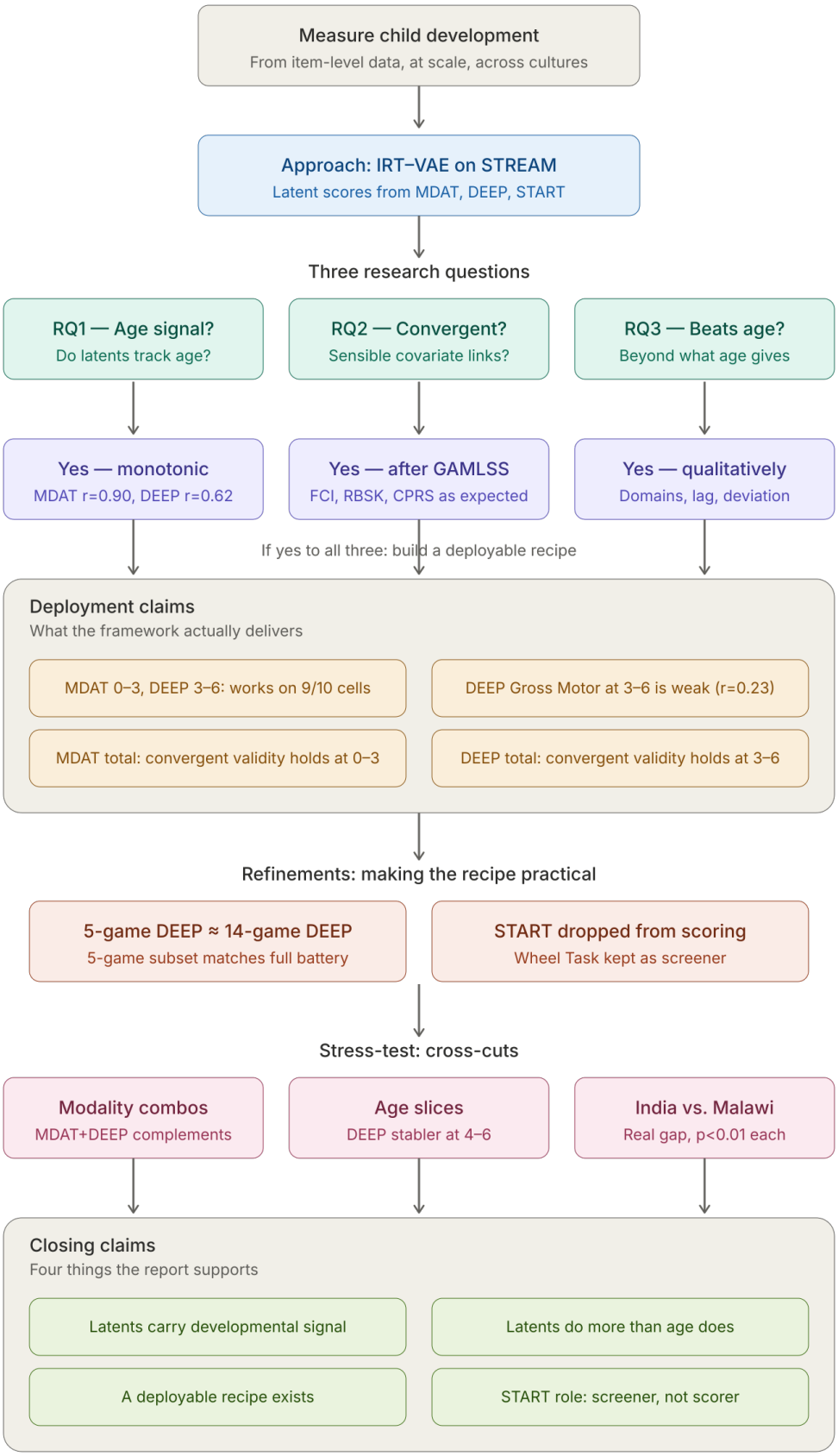


Figure 1.1: Pictorial depiction of the work presented in this report

Chapter 2

Literature Survey

This chapter surveys prior work along the three axes that converge in this report: developmental assessment instruments, item response theory, and deep latent variable models—in particular, hybrid IRT–VAE approaches.

2.1 Developmental Assessment Instruments

Standardised developmental assessment has a long history. The Griffiths Mental Development Scales [Griffiths, 1954], originally published in 1954 and updated as Griffiths-III, score children across five subscales: Foundations of Learning, Language and Communication, Eye and Hand Coordination, Personal–Social–Emotional, and Gross Motor. The Bayley Scales of Infant and Toddler Development [Bayley, 2005] are widely used in research and clinical settings up to age three. The Denver Developmental Screening Test II [Frankenburg et al., 1992] provides a brief screening across four domains.

For lower-resource and parent-report contexts, the Ages and Stages Questionnaire (ASQ) [Squires and Bricker, 2009] relies on caregiver report and is widely deployed in primary-care settings. Tools such as the Malawi Developmental Assessment Tool (MDAT) [Gladstone et al., 2010] were specifically built for African settings, addressing cultural assumptions that travel poorly from Western normative samples. Tablet-based digital assessments have emerged in the last decade. The Developmental Assessment on an E-Platform (DEEP) [Bhavnani et al., 2019] is one such tool, recording rich behavioural traces from interactive games. The WHO's Global Scales for Early Development (GSED) [Cavallera et al., 2023] represent the current state of the art in cross-cultural normative scoring, advocating age-appropriate batteries for different developmental windows.

Tools designed specifically for screening (typical vs. atypical) rather than continuous scoring also exist, including instruments for autism risk such as the Modified Checklist for Autism in Toddlers (M-CHAT) [Robins et al., 2014] and tablet-based screeners in the START family.

2.2 Item Response Theory

Item Response Theory (IRT) emerged in the 1950s–60s as a successor to Classical Test Theory [Lord and Novick, 1968]. The key idea is to model the probability of a correct response to an item as a function of an examinee's latent ability and a small set of item parameters [Baker and Kim, 2004]. The 1-Parameter Logistic (Rasch) model [Rasch, 1960] parameterises items by difficulty alone; the 2-Parameter Logistic (2PL) model adds discrimination; the 3-Parameter Logistic adds a guessing parameter. For ordinal items, the Graded Response Model [Samejima, 1997] introduces a sequence of ordered category thresholds.

Classical estimation uses joint or marginal maximum likelihood, often via EM or MCMC. These methods scale poorly to high-dimensional latent spaces and are slow on large datasets. They also assume specific functional forms.

2.3 Variational Autoencoders

The Variational Autoencoder [Kingma and Welling, 2014] combines a probabilistic generative model with amortised variational inference. The encoder is a neural network that outputs the parameters of an approximate posterior over a latent variable; the decoder is a neural network defining the likelihood of the observation given the latent. Training maximises a tractable evidence lower bound (ELBO) via the reparameterisation trick.

VAEs have been applied to images, text, audio, and tabular data. Their advantages—scalability, end-to-end training, amortised inference—come at the cost of interpretability: latent dimensions are typically not aligned with anything meaningful unless the model is explicitly structured to make them so.

2.4 Hybrid IRT–Deep Learning Approaches

Recent work has sought to combine the best of both worlds: VAE-style scalable training with IRT-style interpretability. Curi et al. [2019] and Wu et al. [2020] both replace the VAE's freeform decoder with an IRT-shaped decoder, recovering item difficulty and discrimination as model parameters while training the encoder end-to-end. Tsutsumi et al. [2021] and Converse et al. [2021] extend this line to ordinal data and to hierarchical multi-domain structures.

These models inherit the missing-data robustness of classical IRT (one can mask out unobserved items in the reconstruction loss) and the speed of VAE inference (a single forward pass to score a new respondent). They are the most direct precursors of the framework used in this report.

2.5 Position of This report

This report builds on the hybrid IRT–VAE line of work and applies it to the STREAM data. The primary methodological contribution is not the architecture itself, which is closely related to existing IRT–VAE proposals, but rather the careful end-to-end validation: from age monotonicity to convergent validity (with proper age-adjustment via GAMLSS) to criterion validity against GMDS, and ultimately to a deployment strategy and a per-instrument characterisation. To our knowledge, the cross-country comparison of DEEP across India and Malawi, and the negative result that START features do not *enhance* continuous scoring, are also new.

Chapter 3

Background: The IRT–VAE Framework

This chapter presents the IRT–VAE framework at the level of intuition required to follow the rest of the report. The full mathematical derivation of the VAE—the ELBO, the reparameterisation trick, and the Monte Carlo estimator—is deferred to Chapter A so that this chapter can be read fluently. Readers familiar with VAEs may skim this chapter and consult the appendix on demand.

3.1 Why Combine IRT and VAEs

A child's response pattern $\mathbf{x} \in \{0, 1\}^d$ across d assessment items is high-dimensional, missing-laden, and generated by a small number of underlying abilities. Two strands of method address this:

- *IRT* treats each item as a logistic function of a latent ability, with item-specific difficulty and discrimination parameters. This is interpretable but classical estimation is slow and the model is rigid.
- *VAEs* use a neural-network encoder to amortise posterior inference and a neural-network decoder to reconstruct $\mathbf{x}$. This is fast and flexible but uninterpretable.

The IRT–VAE framework keeps the VAE encoder—fast, neural—and replaces the VAE decoder with the IRT likelihood. The result is a model whose latent representation is interpretable (it is a multidimensional ability), whose item parameters are interpretable (difficulty and discrimination), and whose training is fast.

3.2 Encoder: Approximating the Posterior

The encoder is a 3–4 layer multilayer perceptron with parameters ϕ. Given a child's response vector $\mathbf{x}_i$, it outputs the parameters—mean and variance—of a Gaussian approximate posterior over the latent ability $\mathbf{z}_i$:

$$q_\phi(\mathbf{z}_i \mid \mathbf{x}_i) = \mathcal{N}\Big(\mathbf{z}_i \mid \mu_\phi(\mathbf{x}_i),\, \operatorname{diag} \sigma^2_\phi(\mathbf{x}_i)\Big).$$

Conceptually, the encoder is a fast, learned amortisation of what classical IRT obtains by per-subject EM or MCMC: an estimate of the latent ability given the responses.

3.3 Decoder: An IRT Likelihood, Not a Neural Net

For binary items, the decoder follows the 2-Parameter Logistic (2PL) IRT model. The probability that subject i correctly answers item j is

$$P(X_{ij} = 1 \mid z_i, a_j, b_j) = \sigma\Big(a_j(z_i - b_j)\Big), \tag{3.1}$$

where a_j is the discrimination of item j (how sharply it separates abilities), b_j is its difficulty (the ability level at which $P = 0.5$), and $\sigma(\cdot)$ is the logistic sigmoid. The item parameters $\{a_j, b_j\}_{j=1}^{J}$ are learned as free parameters of the model jointly with the encoder.

This decoder has two consequences. First, the decoder has *no* neural-network capacity, so the encoder must compress all the information needed to reconstruct $\mathbf{x}$ into the latent ability $\mathbf{z}$. Second, the item parameters retain their classical psychometric interpretation: a high a_j indicates a discriminating item; a high b_j indicates a hard item.

3.4 Training Objective

Training maximises the standard VAE evidence lower bound (ELBO), which for one subject reads

$$\mathcal{L}(\mathbf{x}_i) = \mathbb{E}_{q_\phi(\mathbf{z}_i \mid \mathbf{x}_i)}\Big[\log p_\theta(\mathbf{x}_i \mid \mathbf{z}_i)\Big] - \mathrm{KL}\Big(q_\phi(\mathbf{z}_i \mid \mathbf{x}_i) \,\|\, p(\mathbf{z}_i)\Big), \tag{3.2}$$

where $p(\mathbf{z}_i) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the prior. The first term is the reconstruction log-likelihood under the IRT decoder; the second is a regulariser pulling the posterior towards the unit Gaussian.

The expectation in (3.2) is estimated by drawing one (or a small number of) sample(s)

from the encoder using the reparameterisation trick:

$$\mathbf{z}_i^{(s)} = \mu_\phi(\mathbf{x}_i) + \sigma_\phi(\mathbf{x}_i) \odot \boldsymbol{\epsilon}^{(s)}, \qquad \boldsymbol{\epsilon}^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

This makes the whole pipeline differentiable so that gradients flow back through the encoder. The full derivation, including the Monte Carlo estimator and Jensen's inequality, is in Chapter A.

We empirically verified in Stage 1 of this work that this encoder-amortised estimator recovers ability scores essentially identical to classical EM-based IRT (Pearson $r \geq 0.98$ across all MDAT and DEEP domains); the per-domain comparison is reported in Chapter C.

3.5 Ordinal Items: The Graded Response Model

Some items are not binary. DEEP, for instance, records continuous performance metrics (accuracy, timing, level reached). We discretise these into K ordinal categories. For ordinal items, the 2PL is replaced by Samejima's Graded Response Model (GRM) [Samejima, 1997]: each item still has a single discrimination a_j, but it now has $K-1$ ordered thresholds $b_{j1} < \cdots < b_{j,K-1}$. The cumulative probability of being in category k or above is

$$P(X_{ij} \geq k \mid z_i) = \sigma\Big(a_j(z_i - b_{jk})\Big),$$

and the probability of exactly category k is the difference of adjacent cumulatives. As a concrete example, DEEP's "highest level reached" feature in a 5-level game has K=5 and four thresholds, capturing the increasing ability needed to advance through levels.

The training objective is unchanged in structure: replace the Bernoulli log-likelihood in the reconstruction term with the categorical log-likelihood under the GRM.

3.6 Handling Missing Responses

Real data has missingness. In MDAT, items below a child's age-appropriate starting point are marked correct by convention; items above the ceiling (after several consecutive failures) are marked incorrect; items in between are observed. Many DEEP and START features can be missing because a child did not engage with a game or the system did not record cleanly.

We introduce a binary mask M_{ij}:

$$M_{ij} = \begin{cases} 1 & X_{ij} \text{ observed} \\ 0 & X_{ij} \text{ missing} \end{cases}$$

and multiply each item's contribution to the reconstruction loss by M_{ij}:

$$\mathcal{L}_{\text{rec}}(\mathbf{x}_i) = \mathbb{E}_{q_\phi}\left[\sum_{j=1}^{J} M_{ij}\Big(X_{ij}\log\hat{X}_{ij} + (1 - X_{ij})\log(1 - \hat{X}_{ij})\Big)\right]. \tag{3.3}$$

This is principled: missing items contribute neither to the gradients with respect to ϕ nor to those with respect to the item parameters. We do not impute. At inference time, the encoder simply ignores missing items and bases its posterior on the observed ones.

Chapter 4

Datasets

This chapter describes the data that drives the rest of the report. Section 4.1 gives an overview of the STREAM initiative. Sections 4.2–4.4 describe each of the three primary instruments—MDAT, DEEP, and START—in detail. Section 4.5 introduces the convergent variables (HAZ, SES, FCI, MORS, CPRS, RBSK, CPAS) used to validate the latent scores. Section 4.6 introduces the GMDS as the criterion (clinical gold standard) instrument.

4.1 The STREAM Initiative

The data used in this report was collected under the STREAM (Screening Tools for Reaching Every child to Assess and Monitor) project, an international collaboration that brought together three complementary assessment instruments—MDAT, DEEP, and START—together with anthropometric, socio-economic, and psychosocial measures, and a clinical reference (GMDS). The integrated dataset spans approximately 4,000 subjects across roughly 1,000 features. Of these subjects, approximately 3,700 are prototypically neurotypical; the remaining children are those flagged for further evaluation.

4.2 MDAT: Malawi Developmental Assessment Tool

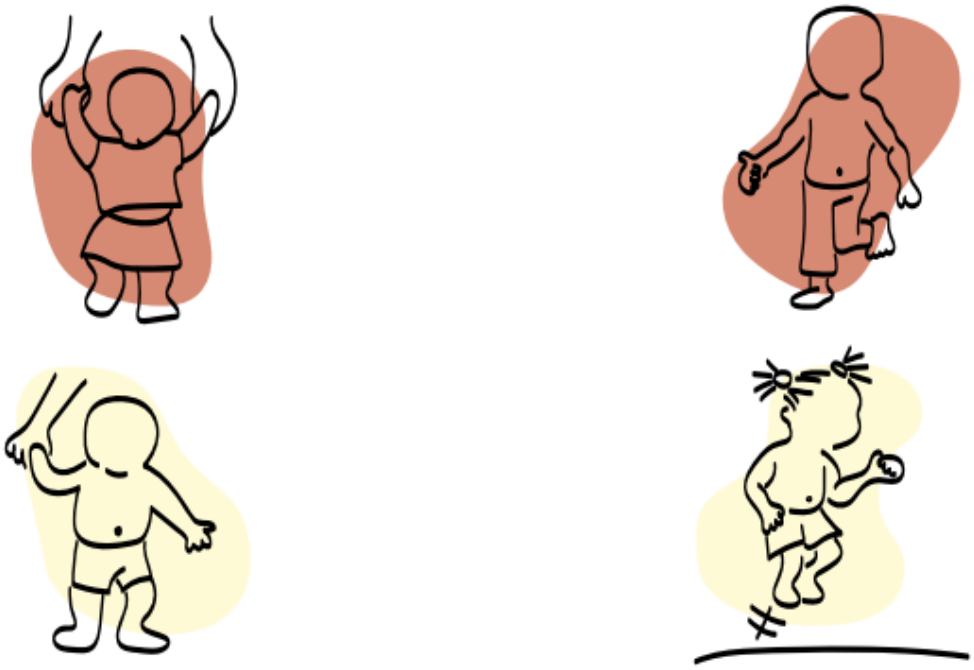


Figure 4.1: MDAT tasks example. The instrument is clinician-administered: each task is presented to the child in a standardised way and scored as success / failure on a paper protocol.

MDAT [Gladstone et al., 2010] is a clinician-administered developmental assessment originally validated in rural Malawi and now widely used across low- and middle-income settings. It evaluates the child across four developmental domains. We summarise each below with examples of the kind of tasks asked at different ages.

Social (36 tasks). Tasks span from early infant social behaviours (smiles in response to caregiver, recognises a familiar face) to toddler-stage skills (engages in pretend play, follows two-step instructions, plays cooperatively with another child).

Fine motor (42 tasks). Tasks include grasping a small object with the whole hand, transferring an object hand-to-hand, building a tower of cubes, copying a vertical line, and threading beads.

Gross motor (40 tasks). Tasks span from holding the head up in prone position, to sitting unsupported, walking unaided, kicking a ball, jumping with both feet, and standing on one foot.

Language (39 tasks). Tasks span from babbling, to first words, to naming common objects, to combining two words, to using prepositions and complex sentences.

A characteristic feature of MDAT administration is the *age-dependent starting point*. A 30-month-old does not start at task 1; she starts at the age-appropriate item and is assumed to have passed all earlier items in that domain (these are coded as success, 1). The test continues until a ceiling is reached—several consecutive failures—after which all later items are coded as failure (0). Items between the start and the ceiling carry the observed responses.

4.2.1 Deriving Latent Scores from MDAT

We train one IRT–VAE per domain, plus one on all items concatenated, to obtain five latent scores per subject:

1. **Social latent**: trained on the 36 social items.
2. **Fine motor latent**: trained on the 42 fine-motor items.
3. **Gross motor latent**: trained on the 40 gross-motor items.
4. **Language latent**: trained on the 39 language items.
5. **Total latent**: trained on all 157 items concatenated.

Items are binary, so we use the 2PL decoder. The mask M records observed/imputed entries; in particular, the implied successes below the starting point are treated as observed-successes, while the implied failures above the ceiling are treated as observed-failures.

4.3 DEEP: Developmental Assessment on an E-Platform

Figure 4.2: DEEP games. The child plays a sequence of 14 short games on a tablet.

DEEP [Bhavnani et al., 2019] is a tablet-based assessment in which the child plays a sequence of 14 short games. The tablet logs rich behavioural traces: response times, accuracy, level reached, error patterns. The platform was developed for the Indian and Malawian cohorts and reduces the dependence on a clinician for administration.

The DEEP team computes a set of *derived* features from the raw game logs (the specific subset used in this study is the same subset distributed by the DEEP team for analysis purposes). To make the analysis tractable and to align item types, we aggregate the derived features into four groups of items per subject:

1. **Highest level reached** (10 items, ordinal). Maximum level attained in each game; four games have a single level and are excluded.

2. **Accuracy score** (multiple items, ordinal after discretisation). Accuracy-related features within each game.

3. **End proportion timer score** (multiple items, ordinal). Timing-related features—how quickly the child completed each segment.

4. **Total score** (one item per subject, ordinal). A summary score combining the above.

4.3.1 Deriving Latent Scores from DEEP

Each of the four groups above gives a separate latent score. Because the underlying features are continuous, we discretise each into K ordinal categories using empirical quantile thresholds and then train a GRM-style IRT–VAE (Section 3.5). Missing entries are masked.

4.4 START: Screening Tool for Autism Risk using Technology

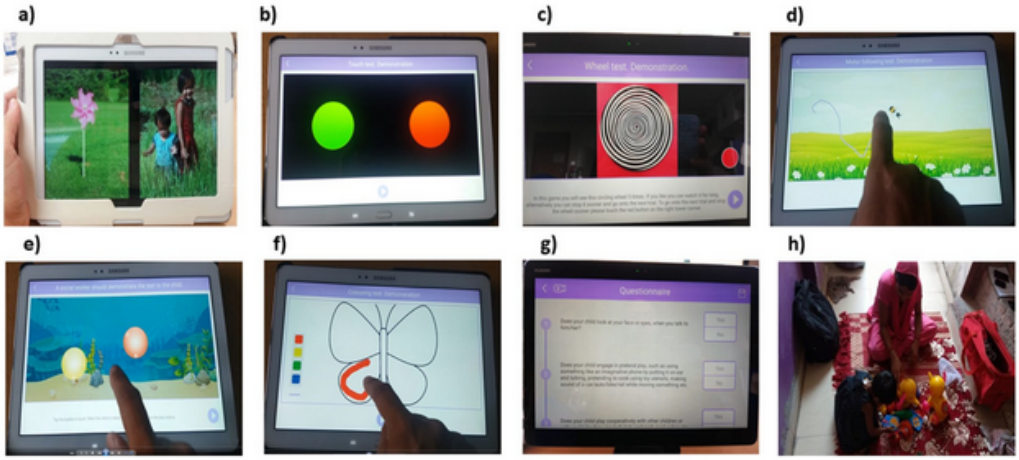


Figure 4.3: Representative tasks from the START battery.

START is a tablet-based instrument originally designed to screen for autism risk and, more broadly, to distinguish typical from atypical neurodevelopment. It comprises eight task families, with features jointly engineered by the IIT Bombay team and the Reading team from the original START project. Examples of tasks include:

- **Wheel Task (WT):** the child interacts with a rotating wheel; features include movement smoothness and engagement.
- **Bubble Popping (BP):** the child pops bubbles on the screen; features include accuracy and reaction time.
- **Preferential Looking Task (PLT):** measures the proportion of time the child fixates on social vs. non-social stimuli (e.g., faces vs. geometric patterns).
- **Button Task (BT):** includes a social-preference feature `bt.soc_pref`.
- **Delayed Gratification Task (DGT)** and others.

START features are continuous, with relatively low missingness compared to MDAT and DEEP. We show in Section 7.2 that the START Total latent correlates only weakly with chronological age and adds nothing to MDAT+DEEP for predicting GMDS. We introduce it here for completeness and analyse it separately.

4.5 Convergent Variables

To validate the latent developmental scores, we use a panel of variables known from the developmental literature to associate with healthy or impaired development. We refer to these as *convergent* variables. Each is described below with the kind of items it draws on.

HAZ — Height-for-Age Z-score. A standardised measure of stunting. Child's height is compared to WHO reference growth standards for the same age and sex; the Z-score is the number of standard deviations from the reference median. Lower HAZ indicates chronic undernutrition and is associated with poorer cognitive and motor development. Correlation direction expected with development scores: *positive*.

SES — Socio-Economic Status. A composite index built from a household-level questionnaire covering items such as: household assets (refrigerator, television, vehicle ownership), housing material (type of roof and walls), household head's education, household head's occupation, source of drinking water, and toilet facility. Higher SES is associated with better developmental outcomes. Correlation direction expected with development scores: *positive*.

FCI — Family Care Indicators. A UNICEF-aligned questionnaire about the home caregiving environment, including items such as: number of children's books in the home, number of varieties of toys, frequency of caregiver play activities (singing, telling stories, naming things), and whether the child was left alone in the past week. Higher FCI indicates a richer caregiving environment. Correlation direction expected with development scores: *positive*.

MORS — Mother Object Relation Scale. A validated scale of the mother–infant relationship, used for infants. It has two subscales:

- *MORS Warmth*: positive items such as "my baby smiles at me" or "I enjoy time with my baby". Correlation direction expected with development scores: *positive*.
- *MORS Invasion*: items capturing intrusive or hostile maternal feelings. Correlation direction expected with development scores: *negative*.

CPRS — Child–Parent Relationship Scale. A toddler-stage instrument with two subscales:

- *CPRS Closeness*: items such as "I share an affectionate relationship with my child". Correlation direction expected with development scores: *positive*.

- *CPRS Conflict*: items such as "my child and I always struggle with each other". Correlation direction expected with development scores: *negative*.

RBSK — Rashtriya Bal Swasthya Karyakram risk indicator. A composite risk indicator from the Indian government's child health screening programme, flagging children at elevated risk of neurodevelopmental disorder based on a structured clinical checklist. Higher RBSK indicates more risk; Correlation direction expected with development scores: *negative*.

CPAS — Childhood Psychosocial Adversity Scale. A caregiver-reported measure capturing psychosocial adversity within the household and surrounding environment. The scale includes indicators of harsh parenting and verbal aggression (e.g., shouting, threats, criticism, insults, and physical harshness), household stress and tension, food insecurity and economic hardship, and exposure to unsafe or adverse neighbourhood conditions such as violence, crime, and harassment. Higher CPAS scores indicate greater psychosocial adversity; Correlation direction expected with development scores: *negative*.

4.6 Criterion Variable: GMDS

Figure 4.4: Griffiths Mental Development Scales (GMDS).

The Griffiths Mental Development Scales [Griffiths, 1954] (latest revision: Griffiths-III) are an internationally used clinical instrument for developmental assessment from birth to six years. GMDS scores a child across five subscales:

1. **Foundations of Learning (FoL)** — attention, memory, problem solving.
2. **Language and Communication (LangCom)** — expressive and receptive language.
3. **Eye and Hand Coordination (Eye-Hand)** — visuomotor integration and fine motor skills.
4. **Personal–Social–Emotional (Soc-Per)** — self-care and social interaction.
5. **Gross Motor** — whole-body movement and balance.

GMDS is administered by trained clinicians and yields normative subscale scores. In the STREAM dataset, GMDS is available on a small subsample (a few hundred children per age band). We use GMDS in two ways. First, as a *criterion* (gold standard) for validating our latent scores: a thin neural network maps our latent representations to GMDS subscale scores in cross-validation, and high correlation indicates that the latent scores recover clinically meaningful structure. Second, as a *bridge to interpretability*: once the bridge model is trained, we can predict GMDS subscale scores for the much larger pool of children who have only MDAT/DEEP data, and use those predicted GMDS scores to inspect convergent validity in clinically familiar units.

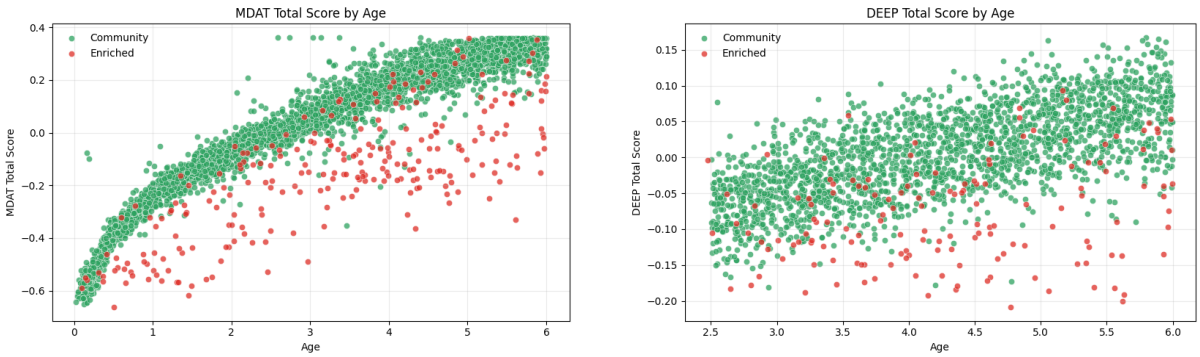
Chapter 5

Validation of Latent Representations

Having defined the IRT–VAE framework (Chapter 3), the data (Chapter 4), and the convergent and criterion variables, we now validate the learned latent scores. The chapter is organised around three questions that any candidate developmental measure must answer.

5.1 Do Latent Measures Track Chronological Age?

A first, basic sanity check: a developmental measure should grow with age. If our latent score is meaningful, the Total latent score should be monotonically increasing in age across the 0–6 year range.



(a) MDAT Total latent score vs. age.

(b) DEEP Total latent score vs. age.

Figure 5.1: Latent Total score against chronological age. Both MDAT and DEEP show the expected positive trend. MDAT grows steadily across the full 0–6 range; DEEP, which is an older-child instrument, shows a flatter early region and steeper growth in 3–6.

Table 5.1 reports the Pearson correlation. Both correlations are large and positive; MDAT, designed for the full 0–6 range and saturated by age 5–6, has the larger correlation.

Table 5.1: Correlation of Total latent scores with chronological age.

Latent Score	***N***	**Correlation with Age**	**95% CI**
MDAT Total	3971	+0.901	(+0.895, +0.907)
DEEP Total	2296	+0.616	(+0.589, +0.641)

Conclusion. Both latent scores increase with age. The lower DEEP correlation reflects DEEP's flatter floor for the youngest children (who cannot meaningfully play tablet games) and ceiling effects in older children. The age signal is strong enough that it must be removed before examining covariates—which we do via GAMLSS in Section 5.2.1.

5.2 Are Latent Measures Sensitive to Adverse Childhood Experiences?

A meaningful developmental measure must respond not only to age but also to known determinants of development—nutrition, socio-economics, caregiving quality. We test this by correlating the latent scores with the convergent variables introduced in Section 4.5, after removing the age trend.

5.2.1 Why Age-Adjustment Is Necessary

Most convergent variables are *family-level*: SES does not change as a child grows, nor does FCI or MORS materially. By contrast, the latent scores rise sharply with age. Suppose two families: family A has higher SES, family B lower. A 2-year-old from family A and a 4-year-old from family B will likely have a lower and higher latent score respectively, despite SES pulling the other way. A naive Pearson correlation between raw latent score and SES is dominated by the age difference and will be positive for the wrong reason—or, in unfortunate cases, negative.

The fix is to regress the latent score on age and use the residual:

$$z_i^{\text{adj}} = z_i - \hat{f}(a_i), \tag{5.1}$$

where $\hat{f}(\cdot)$ is a smooth function of age. A linear regression is too rigid: the latent score is non-linear in age and its variance changes with age (heteroscedasticity). We therefore use GAMLSS (Generalized Additive Models for Location, Scale and Shape) [Rigby and Stasinopoulos, 2005].

GAMLSS

GAMLSS extends linear regression in two ways. First, the conditional mean (location) of the response can be a smooth, nonparametric function of the predictors—typically modelled by penalised cubic splines—rather than a straight line. Second, the conditional standard deviation (scale), and even higher-shape parameters such as skewness and kurtosis, can themselves be smooth functions of the predictors. For our purposes, this means we can write

$$z_i \mid a_i \sim \mathcal{N}\Big(\mu(a_i),\, \sigma^2(a_i)\Big),$$

with both $\mu(\cdot)$ and $\log \sigma(\cdot)$ smooth in age a_i, and obtain $z_i^{\text{adj}} = (z_i - \hat{\mu}(a_i))/\hat{\sigma}(a_i)$ as a properly standardised, homoscedastic residual. This standardised residual is what we correlate with covariates. (Full details in Chapter B.)

5.2.2 Convergent Validity Results

Table 5.2 reports the correlations of the age-adjusted Total latent scores with all convergent variables, separately for MDAT and DEEP across the full 0–6 range. MORS is available only for infants and CPRS only for toddlers; the corresponding cells are blank.

Table 5.2: Convergent validity: correlation of age-adjusted Total latent scores with convergent variables.

	MDAT Total		**DEEP Total**	
Feature	***N***	**Correlation**	***N***	**Correlation**
HAZ	3279	$+0.22^{***}$ (+0.19,+0.25)	1616	$+0.15^{***}$ (+0.10,+0.20)
SES	3965	$+0.05^{**}$ (+0.02,+0.08)	2293	$+0.12^{***}$ (+0.08,+0.16)
FCI	3279	$+0.23^{***}$ (+0.20,+0.26)	2296	$+0.20^{***}$ (+0.16,+0.24)
MORS Warmth	1930	$+0.15^{***}$ (+0.10,+0.19)		
CPRS Closeness			1875	$+0.33^{***}$ (+0.28,+0.37)
MORS Invasion	1930	-0.04 (-0.08,+0.01)		
CPAS	3838	-0.04^{*} (-0.07,-0.00)	1149	-0.04 (-0.10,+0.02)
RBSK	3886	-0.49^{***} (-0.52,-0.47)	2293	-0.39^{***} (-0.43,-0.36)
CPRS Conflict			1875	-0.11^{***} (-0.16,-0.07)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

Conclusion. The directions of all significant correlations match expectations. Variables expected to be positive (HAZ, SES, FCI, MORS Warmth, CPRS Closeness) come out positive; variables expected to be negative (MORS Invasion, CPAS, RBSK, CPRS Conflict) come out negative. RBSK is the strongest single signal: -0.49 on MDAT, -0.39 on DEEP. CPAS, a child-violence-exposure indicator, is small but in the right direction on

both instruments. MORS Invasion against MDAT is the noisiest cell ($r = -0.04$, n.s.); the literature is consistent on MORS Invasion being a weaker subscale than MORS Warmth. The latent scores respond to nutrition (HAZ), socio-economic context (SES, FCI), caregiving quality (MORS, CPRS), violence exposure (CPAS), and aggregate clinical risk (RBSK).

5.3 Is Chronological Age Itself a Sufficient Measure?

Given that age and the latent scores correlate at $r = 0.90$ (Table 5.1), one could ask: why bother building a latent measure when age is already a strong proxy and trivially available? This view is wrong for several reasons that compound.

Age does not differentiate domains. A child who is age-typical on motor skills but well behind on language has the same age as her peers. The whole point of measurement is to reveal this profile—and age, by construction, cannot.

Age does not identify children who are lagging. Two children of the same age can differ enormously in actual ability. Age increases at the same rate for everyone; ability does not. A measure indexed only by age provides no traction on individual variation, which is exactly what is needed to flag a child for intervention.

Age has no cross-country meaning. A 36-month-old in a high-resource setting is, on average, ahead of a 36-month-old in a low-resource setting on most developmental milestones. If a system is calibrated on age alone, the same chronological age means very different things in different contexts. The latent scores, by contrast, are a measurement on a common ability scale; comparison and pooling across cohorts is meaningful.

Age cannot answer "by how much". For policy and intervention, the relevant quantity is not "is this child two years old?" but "by how much, in standardised units, is this child's development below the trajectory expected of her cohort?" That is a residual quantity, computable directly from the latent score after age-adjustment, and not from age itself.

Age is uninformative once developmental ceilings are reached. As a child approaches and surpasses an instrument's ceiling, age continues to grow but ability is no longer being measured. The latent score correctly plateaus at the saturation of the instrument; age does not, leading to spurious "growth" in a regression onto age.

For all these reasons, age is a useful covariate—we use it to build z^{adj}—but cannot replace a dedicated developmental measure. The remaining question, taken up in Chapter 6, is how to make the latent scores *clinically* interpretable, and the answer is GMDS.

Chapter 6

Deployment Strategy and Validity

Having validated that the latent scores carry developmental signal (Chapter 5), we now consider how the framework should be deployed in practice. The key question: do we administer all three instruments to every child?

6.1 An Age-Split Deployment: MDAT for 0–3, DEEP for 3–6

We propose a deployment in which MDAT is administered to children aged 0–3 years and DEEP is administered to children aged 3–6 years. START is treated separately and excluded from the scoring pipeline (Section 7.2). The proposal rests on five practical considerations, elaborated below.

Scalability. Administering MDAT and DEEP to every child doubles assessor time per child, doubles the data-quality and entry burden, and doubles the logistic overhead (consents, devices, room time). At population scale—tens of thousands of children—these costs dominate. Allocating a single instrument per age-band roughly halves the per-child time, which directly translates into either lower cost or larger reach for a fixed budget.

Need for adaptation by age. MDAT items in the 0–3 range include early motor and social milestones (e.g., sitting unsupported, smiling at the caregiver) that simply do not have tablet analogues. DEEP, conversely, requires a child to engage with a tablet and perform structured tasks; pre-3-year-olds typically cannot. The instruments are differentially informative across the age range, and using the more informative instrument for each band is more efficient.

Reduced subjectivity. MDAT scoring depends on the assessor's judgement of whether a task was performed. Inter-rater variability is a documented concern [Gladstone et al.,

2010]. DEEP, in contrast, is automated: timing and accuracy are logged by the tablet without human judgement in the loop. Using DEEP for the 3–6 band reduces subjectivity in exactly the age range where children can meaningfully engage with the device.

Practical training requirements. Training an MDAT assessor takes weeks of supervised practice; certifying a new tablet operator takes hours. Restricting MDAT to 0–3, where it is most informative, reduces the overall training bottleneck for population-scale deployment.

Alignment with international standards. The WHO Global Scales for Early Development (GSED) [Cavallera et al., 2023] explicitly advocate age-stratified assessment batteries, with parent-report or clinician instruments in the youngest band and child-engaged instruments later. Our split mirrors this consensus.

Under this strategy, MDAT latent scores are derived only from children aged 0–3 and DEEP latent scores only from children aged 3–6. The two are not combined per-child.

6.2 Criterion Validity via GMDS

We establish criterion validity by training, for each instrument, a single hidden-layer neural network mapping latent representations to GMDS subscale scores. Sample sizes are moderate (N=462 for MDAT, N=484 for DEEP), so we use 5-fold cross-validation. Correlations are computed between the held-out predictions and the actual GMDS raw scores. We report the mean correlation and 95% confidence interval across folds. Throughout this chapter, *CV* stands for *cross-validation*.

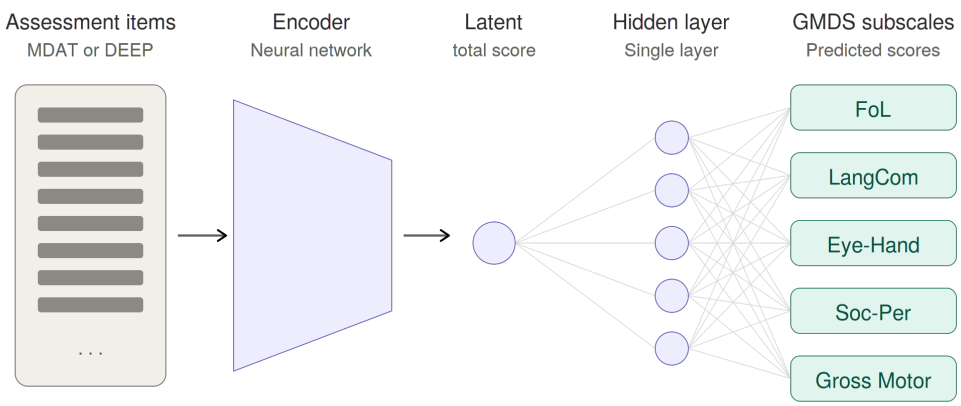


Figure 6.1: Responses are encoded and passed through a Neural Network to map to GMDS (Representative Picture)

Table 6.1 summarises. In the 0–3 band, MDAT predicts every GMDS subscale very

strongly (r between 0.94 and 0.96), with Eye-Hand and Gross Motor narrowly the strongest targets. In the 3–6 band, DEEP predicts each subscale in the moderate-to-strong range (r between 0.58 and 0.81), with Foundations of Learning the strongest and Social-Personal the weakest.

Table 6.1: Criterion validity: 5-fold CV correlation of predicted vs. actual GMDS raw scores. MDAT: $n = 462$; DEEP: $n = 484$.

	MDAT (0–3 yrs)		**DEEP (3–6 yrs)**	
GMDS Target	**Mean Corr.**	**95% CI**	**Mean Corr.**	**95% CI**
FoL	0.95	(0.939, 0.961)	0.81	(0.787, 0.833)
LangCom	0.95	(0.944, 0.956)	0.69	(0.660, 0.720)
Eye-Hand	0.96	(0.949, 0.971)	0.76	(0.721, 0.799)
Soc-Per	0.94	(0.927, 0.953)	0.58	(0.554, 0.606)
Gross Motor	0.96	(0.954, 0.966)	0.63	(0.585, 0.675)

Reading the table. On raw GMDS scores, every subscale-instrument cell clears the bar of useful criterion validity. MDAT correlations in the 0–3 band cluster tightly between 0.94 and 0.96 across all five subscales, with narrow CIs — the instrument tracks the GMDS very closely in this age band. DEEP correlations in the 3–6 band span a wider 0.58–0.81 range; Foundations of Learning is the strongest match and Social-Personal the weakest, though even the latter sits comfortably in the moderate-to-strong band with a tight CI.

6.3 Convergent Validity of Predicted GMDS Scores

Using the bridge networks trained in Section 6.2, we compute predicted GMDS subscale scores for *all* children—not just the GMDS subsample. We then correlate these predicted GMDS scores with the convergent variables. This is the cleanest test of clinical interpretability: do the predicted GMDS scores, in their own clinically familiar units, retain the convergent associations the latent scores had?

6.3.1 Mapping MDAT to GMDS, Ages 0–3

Table 6.2 reports the correlations of the predicted GMDS General score (the average of the five subscales: FoL, LangCom, Eye-Hand, Soc-Per, Gross Motor) with the convergent variables, in the 0–3 age band.

Table 6.2: Convergent validity of the predicted GMDS General score from MDAT (0–3 yrs).

		General
Feature	***N***	**Corr.**
HAZ	1950	-0.21*** (-0.26,-0.17)
SES	1963	-0.01 (-0.05,+0.04)
FCI	1967	+0.09*** (+0.05,+0.14)
MORS Warmth	1926	+0.25*** (+0.21,+0.29)
MORS Invasion	1926	+0.23*** (+0.19,+0.28)
CPAS	1928	+0.13*** (+0.08,+0.17)
RBSK	1884	-0.15*** (-0.19,-0.10)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

Reading this table. The cells expected to be positive (FCI, MORS Warmth) are positive and significant; RBSK is negative and significant. The cells *expected to be negative* but coming out positive—HAZ, MORS Invasion, CPAS—deserve attention. We address this directly in Section 6.4: the predicted GMDS General score inherits the convergence pattern of GMDS itself (computed on the held-out test set), and several of these anomalies are present in GMDS as a measurement instrument, not introduced by the prediction step.

6.3.2 Mapping DEEP to GMDS, Ages 3–6

Table 6.3 repeats the analysis for DEEP in the 3–6 band, again on the General score.

Table 6.3: Convergent validity of the predicted GMDS General score from DEEP (3–6 yrs).

		General
Feature	***N***	**Corr.**
HAZ	1304	+0.15*** (+0.10,+0.20)
SES	1977	+0.07** (+0.02,+0.11)
FCI	1979	+0.17*** (+0.12,+0.21)
CPRS Closeness	1874	+0.35*** (+0.31,+0.39)
RBSK	1977	-0.40*** (-0.44,-0.36)
CPAS	1875	-0.03 (-0.08,+0.01)
CPRS Conflict	1874	-0.14*** (-0.18,-0.09)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

Reading this table. For the 3–6 band, every signed cell goes the right way: positive determinants (HAZ, SES, FCI, CPRS Closeness) are positive, risk variables (RBSK, CPRS Conflict) are negative. The strongest signals are CPRS Closeness ($r = +0.35$) and RBSK ($r = -0.40$). CPAS is small and not significant; the same is true of CPAS against GMDS itself in the 3–6 cohort (Section 6.4), so the failure is a property of the criterion, not the predictor.

6.4 Convergent Validity of GMDS Itself

The MDAT to GMDS table in Section 6.3.1 shows several cells with unexpected signs (HAZ, MORS Invasion, CPAS) and a near-zero cell (SES). Before attributing this to the prediction model, we have to ask whether the criterion—GMDS itself—shows the expected convergence pattern in this dataset. If GMDS does not, the predicted GMDS scores cannot be expected to.

We therefore compute convergent validity directly between the observed (ground-truth, GAMLSS-age-adjusted) GMDS General score and the convergent variables, on the combined India + Malawi sample of children with GMDS administration. To match the age stratification used for the predictors, we split this analysis into the 0–3 and 3–6 bands.

Table 6.4: Convergent validity of the GMDS General score (combined India + Malawi sample), 0–3 yrs.

	General (0–3 yrs)	
Feature	***N***	**Corr.**
HAZ	467	$+0.30^{***}$ (+0.22,+0.38)
SES	467	+0.10 (+0.01,+0.19)
FCI	468	$+0.19^{***}$ (+0.10,+0.28)
MORS Warmth	447	+0.06 (-0.04,+0.15)
CPRS Closeness	11	+0.23 (-0.43,+0.73)
CPAS	457	-0.13^{*} (-0.22,-0.04)
RBS Risk	459	-0.16^{**} (-0.25,-0.07)
MORS Invasion	447	-0.08 (-0.18,+0.01)
CPRS Conflict	11	-0.06 (-0.64,+0.56)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

Table 6.5: Convergent validity of the GMDS General score (combined India + Malawi sample), 3–6 yrs.

	General (3–6 yrs)	
Feature	***N***	**Corr.**
HAZ	309	$+0.28^{***}$ (+0.18,+0.38)
SES	478	+0.09 (-0.00,+0.18)
FCI	478	$+0.20^{***}$ (+0.11,+0.28)
CPRS Closeness	458	+0.09 (+0.00,+0.18)
CPAS	458	+0.03 (-0.07,+0.12)
RBS Risk	478	-0.18^{***} (-0.26,-0.09)
CPRS Conflict	458	-0.05 (-0.14,+0.04)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

What this tells us. GMDS itself shows the expected convergent pattern only partially in this dataset. In both age bands, the positives (HAZ, FCI) are positive and significant, and RBS Risk is negative and significant. SES is weak in both bands. CPRS Closeness is significantly positive in the 0–3 GMDS predictions but only weakly so against ground-truth GMDS, where the 0–3 sample is too small ($n = 11$) to draw conclusions and the 3–6 estimate ($r = +0.09$) is marginal. CPAS against GMDS is near zero in the 3–6 band and even slightly *negative* in the 0–3 band ($r = -0.13$), neither matching the strong positive signal seen in the MDAT-predicted GMDS table. MORS Invasion, which appeared positive in the MDAT predictions, is negative and non-significant against GMDS itself.

The point of this section is attribution. Where the prediction tables in Sections 6.3.1 and 6.3.2 depart from the literature, those departures partly come from how the convergent variables relate to GMDS in this sample, not solely from the latent to GMDS bridge. The HAZ sign flip and the CPAS/MORS Invasion anomalies on the MDAT side, however, are not reproduced by GMDS itself in the 0–3 band, so for those the prediction step is contributing to the mismatch as well. On the DEEP side (3–6), the predicted-GMDS profile aligns with the GMDS-itself profile reasonably well.

6.5 Convergent Validity of Age-Split Latent Totals

The previous subsection routed everything through GMDS. As a final check, we ask whether the raw age-adjusted Total latent scores—without the GMDS bridge—retain convergent validity when restricted to their respective deployment age-bands. This is the most parsimonious test of the deployment strategy.

6.5.1 MDAT Total, Ages 0–3

Table 6.6: Convergent validity: MDAT Total latent score, age-adjusted, ages 0–3.

Feature	***N***	**Correlation**
HAZ	1950	$+0.15^{***}$ (+0.11,+0.19)
SES	1963	+0.03 (-0.01,+0.08)
FCI	1967	$+0.29^{***}$ (+0.25,+0.33)
MORS Warmth	1926	$+0.17^{***}$ (+0.13,+0.21)
MORS Invasion	1926	-0.04 (-0.08,+0.01)
CPAS	1928	-0.06^{**} (-0.11,-0.02)
RBSK	1884	-0.41^{***} (-0.44,-0.37)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

6.5.2 DEEP Total, Ages 3–6

Table 6.7: Convergent validity: DEEP Total latent score, age-adjusted, ages 3–6.

Feature	***N***	**Correlation**
HAZ	1304	$+0.15^{***}$ (+0.10,+0.20)
SES	1977	$+0.11^{***}$ (+0.07,+0.15)
FCI	1979	$+0.22^{***}$ (+0.18,+0.26)
CPRS Closeness	1874	$+0.33^{***}$ (+0.29,+0.37)
RBSK	1977	-0.41^{***} (-0.45,-0.38)
CPAS	1875	-0.02 (-0.07,+0.02)
CPRS Conflict	1874	-0.11^{***} (-0.16,-0.07)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. 95% confidence intervals in parentheses.

Why this section is here, not elsewhere. The age-split convergent analysis closes a loop opened in Section 5.2. There, we showed convergent validity for the full 0–6 range using both instruments. Here we show that, when restricted to their respective deployment bands, each instrument's Total still independently exhibits convergent validity. In particular, RBSK shows strong negative correlations in both bands ($r = -0.41$ in each), and FCI is positively correlated in both ($r = +0.29$ and $r = +0.22$). The deployment strategy does not rely on the GMDS bridge to exhibit convergent validity; the underlying latent scores retain it on their own.

6.6 Summary of the Deployment Strategy

The age-split deployment is supported by four pieces of evidence: practical and theoretical justifications for the split itself (Section 6.1); high criterion validity against GMDS in each band (Section 6.2); convergent validity of the predicted GMDS scores (Section 6.3), with the discrepancies traced to GMDS itself (Section 6.4); and convergent validity of the age-adjusted Totals within each band (Section 6.5). Two practical questions remain: DEEP takes around three hours per child, which is impractical at scale, and the third instrument START has been kept out of the pipeline so far without justification. We address both in the next chapter.

Chapter 7

Operational Refinements: Reducing DEEP and START for Screening

The deployment proposed in Chapter 6 works statistically. This chapter addresses the operational gaps. Section 7.1 shortens DEEP from a 14-game, three-hour battery to a 5-game subset that matches it on GMDS criterion validity. Section 7.2 explains why START is not part of the deployment, by passing it through a continuous-response IRT–VAE and showing it fails on the most basic developmental check (correlation with age) and adds nothing to the MDAT+DEEP scoring pipeline.

7.1 Reducing DEEP Administration Time via Game Subset Selection

DEEP takes approximately three hours per child to administer. For a tool meant to operate at scale—particularly in low-resource clinical settings—this is the single biggest practical constraint. The question we ask here is: *can a subset of DEEP's 14 games preserve criterion validity against GMDS?* If yes, the deployment becomes meaningfully more practical.

7.1.1 The Problem and the Combinatorial Wall

There are $2^{14} = 16{,}384$ possible non-empty subsets of DEEP's 14 games. The naive evaluation strategy is to retrain a separate IRT–VAE on each subset, derive latents, regress to GMDS, and pick the best. Even at a modest training cost per VAE, this is prohibitively expensive and conceptually wasteful: most of the 16,384 jobs differ by a single game and re-learn near-identical structure.

7.1.2 Decomposition: Singleton Latents Plus a Combining Network

We decouple the expensive step (IRT–VAE training) from the combinatorial step (subset selection):

1. **Train 14 singleton IRT–VAEs.** For each game $g \in \{1, \ldots, 14\}$, train a separate IRT–VAE using only that game's items. Each model produces, for every child, the same four latents we use elsewhere in the report: *accuracy*, *end proportion*, *highest level reached*, and *Total*. This step costs 14 VAE training runs and is performed once.

2. **Sweep 16,384 NN regressions.** For any subset $S \subseteq \{1, \ldots, 14\}$, concatenate the per-game latents of games in S to form a $4|S|$-dimensional input. Feed this through a fixed single-hidden-layer neural network (32 units; identical across all subsets to keep capacity controlled), with five output heads—one per GMDS subscale. Train the NN with 5-fold cross-validation, score the held-out correlation per subscale, and average across subscales as the selection metric.

3. **Select.** Rank the 2^{14} candidate NNs and inspect the top-performing subsets.

The expensive step runs 14 times instead of 16,384 times. The combinatorial sweep happens entirely in cheap NN-regression space.

7.1.3 Technical Merits of the Decomposition

The decomposition has three merits beyond raw computational savings.

Modularity. Each game's IRT–VAE is independent of the others. Adding a new game requires training only one new VAE; removing a game requires no retraining. This is impractical with a joint VAE, where every change of game set demands a fresh end-to-end training run.

Interpretability of contribution. The per-game latents are isolable inputs to the combining NN. We can ask which game's latents are doing work in a given prediction, which is opaque in a joint VAE where contributions are entangled in a single shared latent vector.

Reuse of the existing scoring head. The combining NN has the same architecture as the GMDS bridge used in Section 6.2. The only thing that changes between subsets is the input width.

Implicit assumption. The approach assumes per-game singleton latents are sufficient statistics for what each game contributes—that cross-game synergies a joint VAE might capture in its shared latent are recoverable by the combining NN's nonlinearity over per-game latents. Whether this assumption is loss-free is an empirical question we return to in Section 7.1.5.

7.1.4 Results: Five Games Match Fourteen

Table 7.1 compares criterion validity for the full 14-game configuration against a 5-game subset selected by the procedure above.

Table 7.1: Criterion validity for predicting GMDS subscales from DEEP latents: full 14-game battery vs. 5-game subset.

	14 games		**5-game subset**		
GMDS Subscale	**Mean**	**95% CI**	**Mean**	**95% CI**	**Δ**
FoL	0.845	(0.821, 0.869)	0.82	(0.803, 0.837)	−0.03
LangCom	0.752	(0.725, 0.779)	0.75	(0.732, 0.768)	−0.00
Eye-Hand	0.809	(0.774, 0.844)	0.80	(0.778, 0.822)	−0.01
Soc-Per	0.669	(0.644, 0.694)	0.69	(0.679, 0.701)	+0.02
Gross Motor	0.711	(0.690, 0.732)	0.73	(0.712, 0.748)	+0.02
Mean	0.757	—	0.758	—	+0.00

The differences are within the 95% CIs on every subscale. The 5-game subset matches the full 14-game battery on GMDS prediction. The reported subset is one of several near-optimal combinations from the sweep; the open work below addresses what remains before this can be deployed.

7.1.5 Open Questions

Four things remain before this is a deployable protocol.

Selection-vs-evaluation overfitting. The reported correlations are for a subset chosen by ranking 2^{14} candidates by held-out CV performance. Selection on the same data introduces a positive bias. A clean estimate requires nested cross-validation: an outer fold for evaluation, an inner sweep for selection.

Decomposition fidelity. The full-14 row of Table 7.1 should be compared directly against the original joint-14-game IRT–VAE result in Section 6.2. If the decomposed full-

14 model underperforms the joint-14 model, there is a structural cost to the decomposition that should be quantified separately from the cost of subset selection.

Multiple equivalent subsets. Several subsets perform near-equally to the one reported. Computing the top-K subsets and the per-game appearance frequency would distinguish essential games from interchangeable ones, which matters for real deployments that may need to swap games for clinical or logistical reasons.

Convergent validity of the subset. Criterion validity against GMDS is preserved. We have not yet checked that the subset preserves convergent validity against HAZ, SES, FCI, MORS, CPRS, RBSK, and CPAS the way the all the tasks do (Section 6.5). A subset that matches GMDS but loses convergent signal is a worse deployment.

Real-world administration time. A 5/14 game count is not a 5/14 time saving. Games differ in length, instructions, and child attention demand. Once the equivalent-subset pool is characterised, the operational choice should optimise expected wall-clock time across that pool subject to retaining the validity properties above.

7.2 START: Continuous IRT–VAE and the Age Test

START is the third tablet-based instrument in STREAM, designed originally for autism risk screening. It has been kept out of the scoring pipeline so far. This section justifies that exclusion using the same machinery used to validate MDAT and DEEP. We pass START through a continuous-response IRT–VAE, derive a single Total latent score per child, and apply the most basic developmental check: does the score correlate with chronological age? If a candidate developmental measure does not correlate with age in the 0–6 range, no further validity analysis matters.

7.2.1 Continuous IRT–VAE for START

START's items are continuous (e.g., proportion of time looking at a stimulus, mean reaction time), unlike MDAT's binary items and DEEP's ordinal levels. The 2PL likelihood used for MDAT is therefore inappropriate. We extend the IRT–VAE to a continuous response model.

From binary to continuous. In the binary 2PL formulation used for MDAT, each item j has discrimination $a_j > 0$ and difficulty $b_j \in \mathbb{R}$, and the probability of a correct response given latent ability z_i is

$$P(X_{ij} = 1 \mid z_i) = \sigma\Big(a_j z_i - b_j\Big), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \tag{7.1}$$

The logistic item response function (IRF) in (7.1) maps the latent trait to a Bernoulli observation. With a continuous observation $x_{ij} \in \mathbb{R}$, a Bernoulli likelihood is no longer appropriate. The natural IRT generalisation replaces the logistic IRF with a linear–Gaussian IRF:

$$x_{ij} \mid z_i \sim \mathcal{N}\big(a_j z_i + b_j,\ 1\big). \tag{7.2}$$

The discrimination parameter a_j is now the slope of the IRF (how strongly the item responds to ability), and b_j is its intercept (the expected response at $z_i = 0$). Training maximises the same ELBO as for the binary model, but with the Bernoulli log-likelihood replaced by the Gaussian log-likelihood—which, dropping additive constants, is simply mean squared error between observed and predicted item values. We use this MSE-based formulation for START. The full ELBO derivation parallels the binary case in Chapter A; only the reconstruction term changes.

7.2.2 The Age Test

We test whether the START Total latent score correlates with chronological age. Because START's tasks are not all administered to all ages—only the Preferential Looking Task (PLT) is feasible for children under 2.5 years, while all the 8 tasks are administered from 2.5 onwards—we run the test separately in the two bands, using whichever subset of tasks is available in each.

Younger children (ages 0–2.5). For this band, the START Total is computed from PLT alone. Figure 7.1 and Table 7.2 report the result.

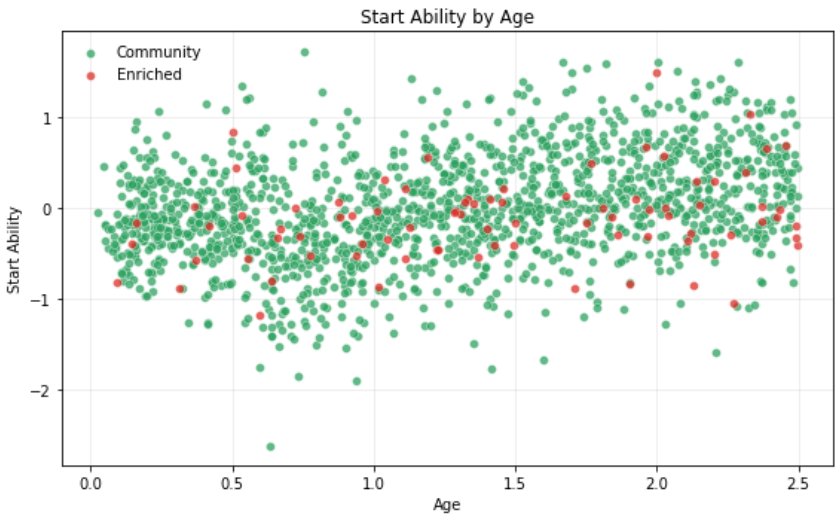


Figure 7.1: START Total latent score (PLT only) against chronological age, ages 0–2.5.

Table 7.2: Correlation of START Total (PLT only) with chronological age, ages 0–2.5.

Latent Score	N	Correlation with Age (95% CI)
START Total (PLT only)	1651	+0.330 (+0.296, +0.381)

Older children (ages 2.5–6). For this band, the START Total is computed from the all the tasks. Figure 7.2 and Table 7.3 report the result.

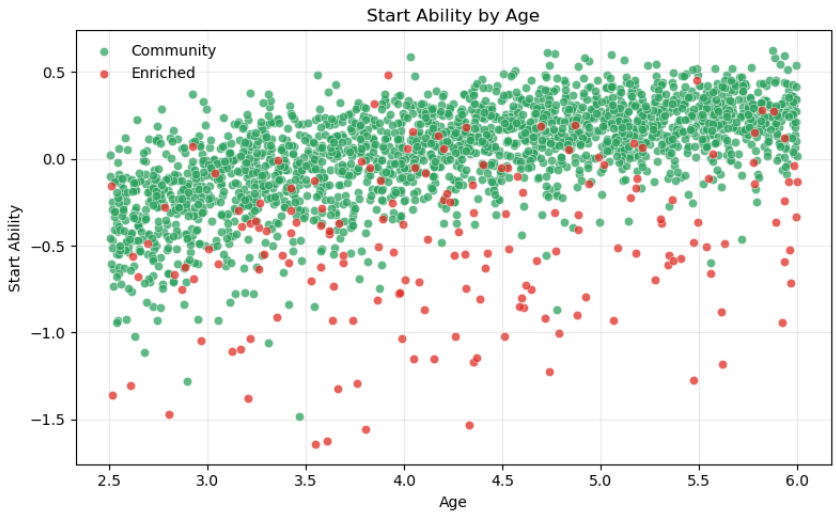


Figure 7.2: START Total latent score (all the tasks) against chronological age, ages 2.5–6.

Table 7.3: Correlation of START Total (all the tasks) with chronological age, ages 2.5–6.

Latent Score	N	Correlation with Age (95% CI)
START Total (all the tasks)	2342	+0.520 (+0.492, +0.550)

Comparison across bands. Both correlations are positive and statistically significant. The 0–2.5 correlation ($r = 0.33$, $N = 1651$, PLT only) is necessarily computed from a thinner feature set; the 2.5–6 correlation ($r = 0.52$, $N = 2342$, all the tasks) is the relevant comparison against MDAT and DEEP, which on the same cohort track age at 0.90 and 0.62 respectively (Table 5.1). START tracks chronological development in both bands, but more weakly than either MDAT or DEEP, including in the older band where the all the tasks are available.

7.2.3 Does START Add Anything to MDAT + DEEP for GMDS?

We compare three configurations on the cohort with all three instruments and a GMDS reference label: START alone, MDAT + DEEP, and MDAT + DEEP + START. All three use the same singleton-plus-NN scoring pipeline (refer to 6.1).

Table 7.4: Predicting GMDS subscales using START alone (N = 949, 5-fold CV).

GMDS Subscale	**Mean Corr.**	**95% CI**
FoL	0.58	(0.550, 0.610)
LangCom	0.59	(0.568, 0.612)
Eye-Hand	0.60	(0.574, 0.626)
Soc-Per	0.60	(0.566, 0.634)
Gross Motor	0.59	(0.550, 0.630)

Table 7.5: Predicting GMDS subscales using MDAT + DEEP, no START (n = 560, 5-fold CV).

GMDS Subscale	**Mean Corr.**	**95% CI**
FoL	0.89	(0.867, 0.913)
LangCom	0.85	(0.826, 0.874)
Eye-Hand	0.89	(0.863, 0.917)
Soc-Per	0.83	(0.812, 0.848)
Gross Motor	0.88	(0.855, 0.905)

Table 7.6: Predicting GMDS subscales using MDAT + DEEP + START (n = 560, 5-fold CV).

GMDS Subscale	**Mean Corr.**	**95% CI**
FoL	0.89	(0.867, 0.913)
LangCom	0.84	(0.814, 0.866)
Eye-Hand	0.89	(0.862, 0.918)
Soc-Per	0.82	(0.800, 0.840)
Gross Motor	0.87	(0.844, 0.896)

The MDAT + DEEP and MDAT + DEEP + START rows match to three significant figures on every subscale. The combining network finds no information in the START latents that is not already in the MDAT and DEEP latents. START adds nothing on top of MDAT + DEEP for predicting GMDS.

7.2.4 Conclusion

START does not behave like a developmental scoring instrument on this cohort. The Total latent correlates with age at 0.52, against 0.90 for MDAT and 0.62 for DEEP.

Conditional on MDAT and DEEP, it adds nothing to GMDS prediction. We exclude it from the scoring pipeline. Whether START has clinical or screening utility outside this scoring problem is discussed next.

7.3 START as a Screening Tool

START's design intent is binary screening: distinguish the small *enriched* subgroup—children with elevated risk for neurodevelopmental disorder—from the larger *community* group. The label is highly imbalanced: enriched children represent only 3–15% of any cohort, depending on the dataset intersection used. We test START on this task in three steps: first as a standalone screener (Section 7.3.2), then with each game used in isolation to see which carry signal (Section 7.3.3), and finally as an augmentation to an MDAT + DEEP screening anchor (Section 7.3.4). This third test is the deployment-relevant one.

7.3.1 Setup: Class Imbalance and SMOTE

All screening experiments follow the same pipeline: median imputation to SMOTE on the training fold to classifier (Logistic Regression, Random Forest, or Gradient Boosting), with an 80/20 stratified split and 5-fold cross-validation on the training portion. We report test-set ROC-AUC alongside held-out CV ROC-AUC, plus enriched-class precision (Enr.P), recall/sensitivity (Enr.R), and F1 (Enr.F1).

With class imbalance this severe, a classifier trained on the raw frequencies would predict the majority class on virtually every example, achieving high accuracy and trivial sensitivity. We address this with the Synthetic Minority Over-sampling Technique (SMOTE) [Chawla et al., 2002].

What SMOTE does. SMOTE generates synthetic minority-class examples by interpolating between existing minority examples in feature space. For each minority point x_i, it finds its k nearest minority neighbours, picks one at random (x_j), and creates a new example $x_{\text{new}} = x_i + \lambda(x_j - x_i)$ for a random $\lambda \in [0, 1]$. This thickens the minority class along its existing manifold. SMOTE is applied *only* to the training fold; the test fold remains in its natural distribution, so reported sensitivity and PPV reflect what the screener would see in deployment.

7.3.2 START Alone: Aggregate Performance

How well does START screen using all the 8 tasks, with no MDAT or DEEP information? Table 7.7 reports the test-set performance for the three classifiers on the START-only cohort (N = 2,034; 66 enriched, 3.2%).

Table 7.7: START-only screening baseline (16 features, no MDAT/DEEP). Best CV ROC-AUC in bold.

Model	ROC-AUC	CV ROC-AUC	95% CI	Enr.P	Enr.R	Enr.F1
LR	0.767	**0.774**	(0.731, 0.817)	0.08	0.62	0.13
RF	0.785	0.754	(0.711, 0.797)	0.10	0.15	0.12
GB	0.715	0.696	(0.630, 0.762)	0.14	0.39	0.20

Aggregate START achieves moderate ROC-AUC (best CV 0.774 with LR) but extremely low enriched F1 (best 0.20 with GB). The screener flags many false positives—useful for triage, not as a standalone tool.

7.3.3 Solo Game Performance

Aggregate performance hides variation across games. We trained classifiers using one game's features at a time on a common population (children with that game completed; $N \approx 2{,}250$–$2{,}310$). Table 7.8 summarises the best-classifier result for four games.

Table 7.8: Solo START game screening performance (best classifier per game). Sens. = sensitivity; PPV = positive predictive value.

Game	Best Model	ROC-AUC	CV ROC-AUC	Sens.	PPV	Enr.F1
`dgt` (Digit)	GB	0.484	0.519	0.71	0.07	0.13
`wt` (Wheel)	LR	0.790	0.728	0.73	0.13	0.22
`bt` (Button)	RF	0.538	0.589	0.48	0.07	0.12
`bpt` (Bubble Pop.)	LR	**0.801**	**0.802**	0.61	0.17	0.27

The story splits sharply by game. `bpt` has the strongest solo signal (CV ROC-AUC 0.802). `wt` is moderate. `bt` is weak, and `dgt` is essentially at chance.

7.3.4 Augmenting MDAT + DEEP with START Games

START in isolation is a weak screener. The deployment-relevant question is different: does adding a START game to MDAT + DEEP improve screening? We answer this with an ablation around an MDAT + DEEP anchor.

Anchor: MDAT + DEEP for screening. We first establish the screening performance of MDAT and DEEP totals alone, on the cohort with both instruments completed ($N = 2{,}627$; 80 enriched). This is the anchor against which START contributions are measured.

Table 7.9: Anchor: MDAT + DEEP totals for screening (no START). Best CV ROC-AUC in bold.

Model	ROC-AUC	CV ROC-AUC	95% CI	Enr.P	Enr.R	Enr.F1
LR	0.794	0.820	(0.791, 0.849)	0.30	0.62	0.41
RF	0.948	0.943	(0.924, 0.962)	0.68	0.89	0.77
GB	0.952	**0.947**	(0.938, 0.956)	0.71	0.88	0.79

The anchor is strong. Gradient Boosting on just two features—the MDAT and DEEP latent totals—achieves CV ROC-AUC 0.947 and enriched F1 0.79, more than three times the F1 of the START-only baseline (0.20). The bulk of the discriminative signal lives in the developmental scores, not in the screening probes.

Per-game incremental contribution. We now add each START game to the anchor and recompute screening performance. Table 7.10 reports the change in key metrics relative to the anchor.

Table 7.10: Change in screening performance when one START game is added to MDAT + DEEP (best of RF/GB). Positive numbers indicate improvement over the anchor.

Game added	ΔROC-AUC	ΔCV ROC-AUC	ΔEnr.F1	Verdict
`dgt` (Digit)	−0.020	+0.019	+0.018	Neutral
`bt` (Button)	−0.003	−0.017	−0.050	**Hurts**
`wt` (Wheel)	**+0.022**	+0.000	**+0.074**	**Helps**
`bpt` (Bubble Pop.)	+0.001	+0.014	+0.050	Marginal

The Wheel Task adds genuine signal. Adding `wt` produces the largest gain: ROC-AUC rises from 0.948 to 0.970 (Random Forest), and enriched F1 from 0.77 to 0.84. The Wheel game's three features—proportion of time, median distance, standard deviation of distance—encode fine motor and timing information that the MDAT and DEEP totals do not summarise. The augmented screener (Table 7.11) reaches 84% PPV at 85% sensitivity, an unusually balanced operating point for a screening task at this base rate.

Table 7.11: MDAT + DEEP + Wheel Task screening performance (N = 2,525, 67 enriched).

Model	ROC-AUC	CV ROC-AUC	95% CI	Enr.P	Enr.R	Enr.F1
LR	0.768	0.798	(0.781, 0.815)	0.26	0.66	0.37
RF	0.970	0.943	(0.926, 0.960)	0.84	0.85	**0.84**
GB	0.971	0.942	(0.921, 0.963)	0.71	0.90	0.80

Other games. `bpt` keeps performance close to the anchor and adds a small F1 gain (+0.050); combined with its strong solo performance, this argues that the bubble-popping motor signal carries some independent information, but most of it is already captured by MDAT and DEEP. `dgt` has essentially no effect on the anchor—consistent with its near-random solo performance. `bt` actively degrades performance (ΔF1 $= -0.050$); `bt.soc_pref` appears to add noise rather than orthogonal signal in the combined model.

7.3.5 Implications for Deployment

Two practical points follow. First, an MDAT + DEEP screener already achieves CV ROC-AUC ≥ 0.94 and enriched F1 ≥ 0.77—without any tablet-game probes. For settings where running START is operationally costly, the developmental-score screener alone is a reasonable choice. Second, the only justified augmentation is adding the Wheel Task, which is short, motorically simple, and improves enriched F1 to 0.84. There is no empirical case for including all 16 START features in a screener; the additional games either fail to help or actively hurt.

7.3.6 Conclusion: A Selective Screener

START's role in the framework is a screener, used selectively. It does not extend the scoring pipeline (Section 7.2), but it is not the right tool for that anyway. As a screener used alone, aggregate START is moderate (CV ROC-AUC 0.774, F1 0.20), with strong variation across games. As a screener combined with MDAT + DEEP, only the Wheel Task adds meaningful signal: enriched F1 rises from 0.77 (anchor) to 0.84.

Chapter 8

Age-Stratified Modality Selection

The deployment strategy of Chapter 6 prescribes MDAT for the 0–3 band and DEEP for the 3–6 band. This chapter justifies that recipe empirically by asking, for each age band, which modality or combination of modalities best predicts GMDS. We use 5-fold cross-validation throughout and report the mean and standard deviation of the per-fold correlation between predicted and held-out GMDS subscales.

The setup is the same as in Chapter 6: each modality is passed through its IRT–VAE to obtain latent representations, which are then fed into a single-hidden-layer neural network (6.1) that predicts GMDS subscales. Where multiple modalities are combined, latents are concatenated that is in combinations like MDAT + DEEP, we combine their latent scores and then send them through the NN layer to predict the GMDS scores.

8.1 Single- and Combined-Modality Performance (Toddlers, 2.5–6)

The first set of experiments holds the toddler age range (2.5–6 years, $N \sim 500$) constant and varies the input modality.

Table 8.1: Single-modality criterion validity.

	MDAT only		**DEEP only**		**START only**	
Target	**Corr.**	**95% CI**	**Corr.**	**95% CI**	**Corr.**	**95% CI**
FoL	0.83	(0.799, 0.861)	0.85	(0.826, 0.874)	0.66	(0.622, 0.698)
LangCom	0.84	(0.821, 0.859)	0.75	(0.723, 0.777)	0.51	(0.444, 0.576)
Eye-Hand	0.86	(0.836, 0.884)	0.81	(0.775, 0.845)	0.63	(0.581, 0.679)
Soc-Per	0.84	(0.808, 0.872)	0.67	(0.645, 0.695)	0.41	(0.361, 0.459)
Gross Motor	0.87	(0.852, 0.888)	0.71	(0.689, 0.731)	0.54	(0.483, 0.597)

MDAT alone is strong and uniform across subscales (0.83–0.87). DEEP alone is competitive on FoL (0.85, slightly above MDAT) and Eye-Hand, but weaker on Soc-Per and Gross Motor. START alone is uniformly worst.

Table 8.2: Combined-modality criterion validity.

	MDAT+DEEP		**MDAT+DEEP+START**		**MDAT+START**	
Target	**Corr.**	**95% CI**	**Corr.**	**95% CI**	**Corr.**	**95% CI**
FoL	0.89	(0.866, 0.914)	0.83	(0.805, 0.855)	0.79	(0.737, 0.843)
LangCom	0.85	(0.830, 0.870)	0.74	(0.714, 0.766)	0.69	(0.648, 0.732)
Eye-Hand	0.89	(0.868, 0.912)	0.80	(0.763, 0.837)	0.77	(0.710, 0.830)
Soc-Per	0.84	(0.820, 0.860)	0.67	(0.619, 0.721)	0.64	(0.589, 0.691)
Gross Motor	0.88	(0.852, 0.908)	0.77	(0.738, 0.802)	0.74	(0.672, 0.808)

Reading the combined results. MDAT+DEEP improves on MDAT alone for FoL, Eye-Hand, and Gross Motor; matches it on LangCom and Soc-Per. The two instruments are complementary on toddlers, with the gain concentrated on cognitive and motor subscales. Adding START to either combination degrades performance on every subscale—consistent with the conclusion in Section 7.2 that START does not contribute scoring signal beyond what MDAT and DEEP already provide.

8.2 The 4–6 Band: Where MDAT Loses Ground

Toddlers are not all alike. We zoom in on the 4–6 age band ($N \sim 330$) to ask how the modalities behave at the older end of the toddler range.

Table 8.3: Criterion validity in the 4–6 age band (5-fold CV).

	DEEP only		**MDAT only**		**MDAT+DEEP**	
Target	**Corr.**	**95% CI**	**Corr.**	**95% CI**	**Corr.**	**95% CI**
FoL	0.73	(0.691, 0.769)	0.62	(0.531, 0.709)	0.74	(0.680, 0.800)
LangCom	0.58	(0.527, 0.633)	0.49	(0.289, 0.691)	0.54	(0.396, 0.684)
Eye-Hand	0.65	(0.615, 0.685)	0.58	(0.439, 0.721)	0.60	(0.486, 0.714)
Soc-Per	0.46	(0.413, 0.507)	0.44	(0.249, 0.631)	0.41	(0.246, 0.574)
Gross Motor	0.45	(0.417, 0.483)	0.61	(0.502, 0.718)	0.58	(0.472, 0.688)

Reading the 4–6 results. MDAT alone performs noticeably worse here than in the broader 2.5–6 range, with mean correlations dropping by 0.15–0.35 and CIs widening

substantially (the LangCom CI for MDAT alone spans 0.29–0.69). MDAT loses both accuracy and stability as children approach its design ceiling. DEEP is more stable: its CIs are tight on every subscale, and DEEP outperforms MDAT on FoL, LangCom, and Eye-Hand. Combining MDAT and DEEP in the 4–6 band does not consistently help; gains appear only on FoL, while CIs widen. At the older end of the toddler range, DEEP is the more reliable single modality, which directly motivates the deployment choice of Chapter 6.

8.3 Infants and Younger Toddlers (0–6 and 0–2.5)

At the other end of the spectrum lies the infant and young-toddler range. PLT (Preferential Looking Task) from START is occasionally proposed as a social signal for very young children, so we test whether MDAT + PLT improves over MDAT alone.

Table 8.4: Criterion validity for the 0–6 age band ($N \sim 950$). PLT is the Preferential Looking Task subset of START.

	MDAT only		**MDAT+PLT**	
Target	Corr.	95% CI	Corr.	95% CI
FoL	0.95	(0.939, 0.961)	0.95	(0.946, 0.954)
LangCom	0.97	(0.964, 0.976)	0.97	(0.966, 0.974)
Eye-Hand	0.97	(0.963, 0.977)	0.97	(0.968, 0.972)
Soc-Per	0.97	(0.964, 0.976)	0.97	(0.967, 0.973)
Gross Motor	0.97	(0.964, 0.976)	0.97	(0.966, 0.974)

Table 8.5: Criterion validity for the 0–2.5 age band ($N \sim 400$). PLT is the Preferential Looking Task subset of START.

	MDAT only		**MDAT+PLT**	
Target	Corr.	95% CI	Corr.	95% CI
FoL	0.94	(0.925, 0.955)	0.94	(0.929, 0.951)
LangCom	0.95	(0.933, 0.967)	0.94	(0.935, 0.945)
Eye-Hand	0.95	(0.925, 0.975)	0.95	(0.933, 0.967)
Soc-Per	0.94	(0.923, 0.957)	0.93	(0.921, 0.939)
Gross Motor	0.95	(0.934, 0.966)	0.95	(0.935, 0.965)

In both bands, MDAT alone already saturates at $r \approx 0.94$–0.97. Adding PLT does not change the mean correlation in any meaningful way, including on Soc-Per where one might have hoped it would help. This is a clean negative result: in the infant range, MDAT contains essentially all the GMDS-relevant information; PLT does not add to it.

8.4 Summary

The picture across age bands is consistent and supports the deployment recipe of Chapter 6. In infancy (0–2.5), MDAT alone saturates the prediction; nothing else needs to be added. On toddlers (2.5–6), MDAT remains the strongest single modality, and MDAT+DEEP provides modest gains on cognitive and motor subscales. At the older end (4–6), MDAT loses both accuracy and stability as children approach its design ceiling, and DEEP becomes the more reliable single modality. START, in any combination, fails to contribute scoring signal beyond what MDAT and DEEP already provide. Together these results justify using MDAT throughout the 0–3 band and DEEP throughout the 3–6 band.

Chapter 9

Robustness: Proxying and Generalisation

Having established in Chapter 8 which modality to use at which age, two further questions probe how far the framework stretches beyond its primary use case. First, can the cheap, automated DEEP stand in for the expensive, examiner-administered MDAT (Section 9.1)? Second, do the results travel across the two STREAM cohorts—India and Malawi—or are they country-specific (Section 9.2)? Both are external-validity questions, and both are addressed with the same 5-fold cross-validation protocol used throughout the previous chapter.

9.1 DEEP as a Proxy for MDAT

We test whether DEEP can substitute for MDAT by predicting MDAT subscale scores from DEEP latents. To make MDAT scores accessible without going through the IRT–VAE on the small GMDS-labelled subsample, we use a simple aggregation for the MDAT target: $+1$ for a success, -1 for a failure, 0 for not attempted. This gives a clean target on the full $N \sim 2200$ children with both DEEP and MDAT.

Table 9.1: Predicting MDAT scores from DEEP latent representations (5-fold CV).

MDAT Subscale	**Mean Corr.**	**95% CI**
Fine motor (fm)	0.73	(0.707, 0.753)
Gross motor (gm)	0.59	(0.560, 0.620)
Social (s)	0.54	(0.519, 0.561)
Language (l)	0.74	(0.721, 0.759)
Total (tot)	0.71	(0.701, 0.719)

Reading the proxy results. DEEP predicts MDAT Total at r=0.71 and is best on Fine Motor and Language ($\sim$ 0.73–0.74). This is consistent with DEEP's design: tablet games inherently exercise fine-motor and language abilities. Social and gross motor are weaker, also consistent with the design—DEEP does not solicit social interaction and is sedentary. The implication is that DEEP can stand in for MDAT in the cognitive and fine-motor space at moderate fidelity, which is precisely what is required for the 3–6 deployment band where MDAT is no longer optimal.

9.2 Cross-Country Generalisation: India vs. Malawi

The STREAM cohorts include both Indian and Malawian children. We re-run the criterion-validity experiment separately for each country to ask whether DEEP behaves the same way in different cultural and linguistic settings.

Table 9.2: DEEP criterion validity, India vs. Malawi (5-fold CV).

	India ($N \sim 286$)		**Malawi** ($N \sim 272$)	
Target	**Corr.**	**95% CI**	**Corr.**	**95% CI**
FoL	0.88	(0.858, 0.902)	0.81	(0.778, 0.842)
LangCom	0.83	(0.806, 0.854)	0.72	(0.682, 0.758)
Eye-Hand	0.86	(0.841, 0.879)	0.74	(0.709, 0.771)
Soc-Per	0.79	(0.752, 0.828)	0.62	(0.575, 0.665)
Gross Motor	0.81	(0.786, 0.834)	0.73	(0.705, 0.755)

DEEP performs better on every subscale in India than in Malawi. The pooled-cohort performance lies between the two single-country results. The gap is largest on Soc-Per (0.79 vs. 0.62) and LangCom (0.83 vs. 0.72).

9.2.1 Is the Difference Significant?

We test the difference using Fisher's z-transformation. For each subscale, we test

$$H_0 : \ \rho_{\text{India}} = \rho_{\text{Malawi}} \quad \text{vs.} \quad H_1 : \ \rho_{\text{India}} > \rho_{\text{Malawi}}.$$

The transformation $z = \frac{1}{2}\ln\Big((1+r)/(1-r)\Big)$ approximately normalises the sampling distribution, with standard error $\sqrt{1/(n_1{-}3) + 1/(n_2{-}3)}$.

Table 9.3: Fisher's z test of the India vs. Malawi difference, one-sided.

Target	**Z**	**p-value**
FoL	2.92	0.0017
LangCom	3.29	0.0005
Eye-Hand	4.03	0.00003
Soc-Per	4.07	0.00002
Gross Motor	2.33	0.0099

Every subscale shows a statistically significant difference at $p < 0.01$, robust to a Bonferroni correction over the five subscales. The cross-country gap is real, not noise.

9.2.2 What Drives the Gap?

A statistically significant gap does not specify the cause. Plausible drivers include: (i) game content and instructions that translate or culturally calibrate better in one cohort than the other; (ii) differences in tablet familiarity, prior screen exposure, and engagement, which would inflate measurement noise in the less-familiar cohort; (iii) cohort-level distributional shifts in the underlying GMDS scores themselves, which change the variance available for the predictor to explain; and (iv) sampling differences in the GMDS-labelled subsample, since the two cohorts were not matched on age or socio-demographic composition.

Disentangling these is left for future work. The methodological message stands regardless of which mechanism dominates: DEEP-based scores are not country-invariant, and the framework should be re-validated whenever it is deployed in a new cultural or linguistic context. A model trained on one cohort cannot be assumed to perform at the same level on another, even when the underlying instrument and target are nominally the same.

9.3 Summary

Both probes return useful but bounded results. DEEP can serve as a moderate-fidelity proxy for MDAT, with the strongest performance on the subscales that match DEEP's design (fine motor and language) and weaker performance on subscales it was not built to elicit (social and gross motor). And DEEP generalises better in India than in Malawi, with a statistically significant gap on every subscale. Together these define the limits of the framework: it can be cheapened by using DEEP in place of MDAT where the target is cognitive, but it cannot be assumed to transport across cohorts without re-validation.

Chapter 10

Conclusion

10.1 Summary

This report presents a data-driven framework for measuring early-childhood development that combines IRT with VAEs. The framework is applied to the STREAM dataset, comprising MDAT, DEEP, and START across about 4,000 children. The findings:

Latent scores carry developmental signal. Both MDAT and DEEP latent Total scores are monotonic in chronological age (Section 5.1). After GAMLSS age-adjustment (Section 5.2.1), they correlate in the expected directions with HAZ, SES, FCI, MORS, CPRS, CPAS, and RBSK (Section 5.2).

Latent scores beat age as a developmental measure. Age cannot differentiate domains, identify lagging children, account for cross-country variation, or quantify deviation from a cohort-expected trajectory (Section 5.3). The latent scores can.

Criterion validity against GMDS supports a deployable recipe. A neural-network bridge from latent scores to GMDS subscales predicts at $r \geq 0.94$ for MDAT (0–3) and $r \in [0.58, 0.81]$ for DEEP (3–6), with the predicted GMDS scores retaining convergent validity in each band (Chapter 6). Where the predicted GMDS scores show unexpected signs against HAZ, MORS Invasion, and CPAS, the same pattern is present in GMDS itself on this sample (Section 6.4); the bridge inherits the criterion's behaviour rather than introducing an artefact.

DEEP can be made administratively practical. A decomposed pipeline—14 per-game IRT–VAEs feeding a shared 32-unit combining network—turns the 2^{14}-subset search into a sweep of cheap NN regressions instead of VAE retrainings (Section 7.1). A 5-game subset matches the full 14-game battery on every GMDS subscale (mean correlation 0.758 vs. 0.757), cutting DEEP administration from ∼3 hours toward ∼1 hour.

START is excluded from the scoring pipeline. A continuous-response IRT–VAE on START's items produces a Total latent that correlates with age at only $r = 0.31$ (vs. 0.90 for MDAT and 0.61 for DEEP). Conditional on MDAT and DEEP, START contributes nothing to GMDS prediction: the MDAT+DEEP and MDAT+DEEP+START rows match to three significant figures on every subscale (Section 7.2). Both checks point the same way and START is dropped from the scoring pipeline.

Where each instrument is strongest. MDAT and DEEP are complementary on cognitive subscales for toddlers; DEEP is more stable than MDAT in 4–6 (**??**). DEEP performs better on every GMDS subscale in India than in Malawi, a gap significant by Fisher's z-test on every subscale.

Computational efficiency. Training runs on commodity hardware. The encoder amortises inference, so scoring a new child is a single forward pass.

10.2 Limitations

GMDS is the criterion against which the bridge is trained, and GMDS is itself imperfect on this sample—several convergent variables (SES, CPAS, MORS Invasion) show weak or wrong-signed correlations with GMDS itself, and the prediction inherits this. The IRT–VAE assumes population homogeneity; atypical children are a small minority and may be poorly represented in the latent geometry. The India–Malawi performance gap is documented but not explained. The framework is cross-sectional and does not model individual developmental trajectories.

10.3 Future Work

Completing the DEEP subset analysis. The decomposition opens several follow-ups: nested-CV bias correction, a top-K subset characterisation with per-game appearance frequency, convergent-validity verification of the chosen subset, and a decomposed-14 vs. joint-14 IRT–VAE comparison to quantify the cost of the decomposition itself. *Longitudinal extensions.* A recurrent or state-space variant of the IRT–VAE can model trajectories rather than snapshots, giving measures of developmental velocity and acceleration. *Semi-supervised bridges.* The GMDS subsample can be used inside the model rather than as a downstream regression target, tying the latent space directly to clinical scales. *Domain adaptation across countries.* The India–Malawi gap motivates explicit domain-adaptation losses or country-specific item-parameter shifts. *Other domains of mental health.* The general principle—unsupervised IRT-shaped representation learning over item-level data—should transfer to instruments measuring depression, anxiety, or

behavioural disorders, which face the same missingness, item heterogeneity, and interpretability constraints.

Appendix A

VAE Mathematical Derivations

This appendix expands the high-level treatment of Chapter 3 into the full derivation of the VAE training objective. We follow the standard development of Kingma and Welling [2014].

A.1 The Intractable Posterior

Our aim is the posterior over latent variables z given observed data x:

$$p(z \mid x) = \frac{p(x \mid z)p(z)}{\int p(x \mid z)p(z)\, dz}, \tag{A.1}$$

where $p(z) = \mathcal{N}(0, I)$ is the prior and $p(x \mid z)$ is the data likelihood. The marginal in the denominator generally lacks a closed form and is computationally prohibitive in high-dimensional latent spaces.

A.2 Encoder: Variational Approximation

We replace the true posterior with a variational distribution $q_\phi(z \mid x)$ parameterised by an encoder neural network with parameters ϕ. For a Gaussian latent,

$$q_\phi(z \mid x) = \mathcal{N}\Big(z \mid \mu_\phi(x), \operatorname{diag} \sigma_\phi^2(x)\Big). \tag{A.2}$$

A.3 Decoder

The decoder defines $p_\theta(x \mid z)$ with parameters θ. In the standard VAE the decoder is a neural network; in our IRT–VAE setup it is the 2PL or GRM IRT model, with $\theta = \{a_j, b_j\}_{j=1}^J$.

A.4 Derivation of the ELBO

The marginal log-likelihood admits the standard variational lower bound:

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x \mid z) p(z)\, dz \\
&= \log \int q_\phi(z \mid x)\, \frac{p_\theta(x \mid z) p(z)}{q_\phi(z \mid x)}\, dz \\
&= \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(x \mid z) p(z)}{q_\phi(z \mid x)}\right] \\
&\geq \mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(x \mid z) p(z)}{q_\phi(z \mid x)}\right] \quad \text{(Jensen)} \\
&= \mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)] - \mathrm{KL}\Big(q_\phi(z \mid x) \,\|\, p(z)\Big) \\
&= \mathcal{L}_{\mathrm{ELBO}}(x; \theta, \phi).
\end{aligned} \tag{A.3}$$

The first term is the expected reconstruction log-likelihood; the second is the KL divergence between the variational posterior and the prior.

For a dataset of N subjects:

$$\mathcal{L}_{\mathrm{VAE}} = \sum_{i=1}^{N} \Big[\mathbb{E}_{q_\phi(z_i \mid x_i)}[\log p_\theta(x_i \mid z_i)] - \mathrm{KL}\Big(q_\phi(z_i \mid x_i) \,\|\, p(z_i)\Big)\Big]. \tag{A.4}$$

A.5 Reparameterisation Trick

The expectation $\mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)]$ depends on ϕ through the sampling distribution, and gradients ∇_ϕ cannot pass through the sampling operation directly. We rewrite the latent variable as a deterministic function of an auxiliary noise variable:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{A.5}$$

Now stochasticity is isolated in ϵ, which does not depend on ϕ, and z is a differentiable function of ϕ. The expectation becomes

$$\mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}\Big[\log p_\theta(x \mid \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)\Big]. \tag{A.6}$$

A.6 Monte Carlo Estimator

We approximate the expectation with S samples [Robert and Casella, 2004]:

$$\mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)] \approx \frac{1}{S} \sum_{s=1}^{S} \log p_\theta(x \mid z^{(s)}), \qquad z^{(s)} = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon^{(s)}. \tag{A.7}$$

Kingma and Welling [2014] show that S=1 per data point typically suffices in practice.

A.7 IRT–VAE Objective: 2PL

For the binary IRT–VAE, the per-subject ELBO becomes

$$\begin{aligned}\mathcal{L}(x_i) = \mathbb{E}_{q_\phi(z_i|x_i)}\Bigg[\sum_j X_{ij}\log\hat{X}_{ij} + (1-X_{ij})\log(1-\hat{X}_{ij})\Bigg] \\ - \mathrm{KL}\Big(q_\phi(z_i \mid x_i)\,\|\,p(z_i)\Big),\end{aligned} \tag{A.8}$$

with $\hat{X}_{ij} = \sigma(a_j(z_i - b_j))$. Combined with the reparameterisation trick:

$$\begin{aligned}\mathcal{L}(x_i) \approx \frac{1}{S}\sum_{s=1}^{S}\Bigg[\sum_j X_{ij}\log\hat{X}_{ij}^{(s)} + (1-X_{ij})\log(1-\hat{X}_{ij}^{(s)})\Bigg] \\ - \mathrm{KL}\Big(q_\phi(z_i \mid x_i)\,\|\,p(z_i)\Big),\end{aligned} \tag{A.9}$$

where $z_i^{(s)} = \mu_\phi(x_i) + \sigma_\phi(x_i) \odot \epsilon^{(s)}$ and $\hat{X}_{ij}^{(s)} = \sigma(a_j(z_i^{(s)} - b_j))$. Item parameters $\{a_j, b_j\}$ are updated as free parameters; encoder parameters ϕ via backpropagation through the reparameterisation trick.

A.8 IRT–VAE Objective: GRM (Ordinal)

For ordinal items (Section 3.5), the cumulative probability is

$$P(X_{ij} \geq k \mid z_i) = \sigma(a_j(z_i - b_{jk})), \qquad k = 1, \ldots, K-1, \tag{A.10}$$

with $b_{j1} < b_{j2} < \cdots < b_{j,K-1}$. The probability of exactly category k is

$$P(X_{ij} = k \mid z_i) = P(X_{ij} \geq k \mid z_i) - P(X_{ij} \geq k+1 \mid z_i), \quad k = 1, \ldots, K-1, \tag{A.11}$$

$$P(X_{ij} = K \mid z_i) = P(X_{ij} \geq K \mid z_i). \tag{A.12}$$

The reconstruction term replaces the Bernoulli log-likelihood with the categorical log-likelihood:

$$\log P(X_{ij} \mid z_i, a_j, \{b_{jk}\}) = \sum_{k=1}^{K} \mathbf{1}[x_{ij} = k]\,\log P(X_{ij} = k \mid z_i). \tag{A.13}$$

The full ELBO for one subject:

$$\begin{aligned}\mathcal{L}(\theta, \phi; x_i) = \mathbb{E}_{z_i \sim q_\phi}\left[\sum_{j=1}^{J}\sum_{k=1}^{K} \mathbf{1}[x_{ij} = k] \log P(X_{ij} = k \mid z_i, a_j, b_{jk})\right] \\ - \mathrm{KL}\Big(q_\phi(z_i \mid x_i) \,\|\, p(z_i)\Big).\end{aligned} \tag{A.14}$$

A.9 Masked Reconstruction

With the mask $M_{ij} \in \{0, 1\}$ (Section 3.6), the binary reconstruction term becomes

$$\mathcal{L}_{\mathrm{rec}}(x_i) = \mathbb{E}_{q_\phi}\left[\sum_{j=1}^{J} M_{ij}\Big(X_{ij} \log \hat{X}_{ij} + (1 - X_{ij}) \log(1 - \hat{X}_{ij})\Big)\right]. \tag{A.15}$$

The KL term is unaffected. The same masking applies to the GRM objective. Missing items contribute zero gradient and zero loss; no imputation is required.

Appendix B

GAMLSS: Background

This appendix provides background on GAMLSS [Rigby and Stasinopoulos, 2005], the framework used in Section 5.2.1 for age-adjustment.

B.1 Motivation

A standard linear regression $y = \beta_0 + \beta_1 x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ assumes the conditional mean is linear in x and the conditional variance is constant. Both assumptions fail for developmental data: latent scores rise sharply early in life and plateau; their dispersion changes with age, generally narrowing as instruments saturate.

B.2 The GAMLSS Family

A GAMLSS model writes the conditional distribution of y given covariates as

$$y \mid x \sim \mathcal{D}(\mu(x), \sigma(x), \nu(x), \tau(x)),$$

where $\mathcal{D}$ is a chosen family (Gaussian, t, Box-Cox, etc.) and each parameter is itself a smooth function of the covariates:

$$g_1(\mu(x)) = \eta_1(x), \quad g_2(\sigma(x)) = \eta_2(x), \quad \ldots$$

with g_k link functions and η_k additive predictors that may include penalised cubic splines and other smoothers. The full model is fit by penalised maximum likelihood.

B.3 Use in This report

For each latent score, we fit the simplest useful GAMLSS:

$$z_i \mid a_i \sim \mathcal{N}(\mu(a_i), \sigma^2(a_i)),$$

with $\mu(a)$ a penalised cubic spline in age and $\log \sigma(a)$ a (typically lower-degree) penalised cubic spline. The age-adjusted score is the standardised residual

$$z_i^{\text{adj}} = \frac{z_i - \hat{\mu}(a_i)}{\hat{\sigma}(a_i)},$$

which is approximately mean-zero, unit-variance, and free of the age trend—suitable for direct correlation with covariates.

Appendix C

Stage 1: Empirical Equivalence of the IRT–VAE with Classical IRT

Before adopting the IRT–VAE framework for the analyses in the main report, we verified in Stage 1 of this work that the VAE-based encoder produces latent ability estimates that closely match those of classical IRT estimation. This appendix summarises that validation. Classical IRT parameters were fitted with the `mirt` package in R using marginal maximum likelihood (EM); the IRT–VAE was trained as described in Chapter 3 and Chapter A. For each domain, we then computed the Pearson correlation between the per-subject ability scores produced by the two methods.

C.1 MDAT

We trained one IRT–VAE per MDAT domain (Social, Fine Motor, Gross Motor, Language) and one on all items concatenated (Total), using the 2PL decoder for the binary items.

Table C.1: Correlation between Classical IRT (`mirt`) and IRT–VAE ability scores on MDAT.

Domain	***N***	**Correlation**
Social	3960	0.98
Fine Motor	3956	0.99
Gross Motor	3956	0.98
Language	3956	0.99
Total	3961	0.98

C.2 DEEP

For DEEP, the four item groups (Highest Level Reached, Accuracy, End Proportion Timer, Total) were modelled with the GRM-style ordinal IRT–VAE (Section 3.5) after discretising continuous features into ordinal categories.

Table C.2: Correlation between Classical IRT and IRT–VAE ability scores on DEEP.

Domain	***N***	**Correlation**
Highest Level	2248	0.98
Accuracy	2296	0.99
End Proportion Timer	2289	0.99
Total	2296	0.99

C.3 Takeaway

Across all nine domain/item-group combinations (five MDAT, four DEEP), the IRT–VAE ability scores correlate with classical IRT estimates at $r \geq 0.98$. This establishes that the encoder-amortised inference used throughout the report recovers essentially the same latent abilities as the well-validated classical EM-based estimator, while also providing the scalability, missingness handling, and end-to-end differentiability that the downstream analyses (GMDS bridge, GAMLSS age-adjustment, subset selection) require.

Bibliography

F. B. Baker and S.-H. Kim. *Item Response Theory: Parameter Estimation Techniques.* CRC Press, 2nd edition, 2004.

N. Bayley. *Bayley Scales of Infant and Toddler Development, Third Edition.* Harcourt Assessment, 2005.

S. Bhavnani, G. Mukherjee, A. Kuruvilla, et al. Development and validation of a digital assessment tool for measuring neurodevelopment in children (DEEP). *The Lancet Digital Health*, 1(3):e123–e131, 2019.

V. Cavallera, M. Lancaster, et al. The Global Scales for Early Development (GSED): A WHO-led measurement framework. *Bulletin of the World Health Organization*, 2023.

N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.

G. Converse, S. Curi, and S. Oliveira. Variational item response theory: Fast, accurate, and expressive. *International Conference on Educational Data Mining (EDM)*, 2021.

M. Curi, G. Converse, J. Hajewski, and S. Oliveira. Interpretable variational autoencoders for cognitive models. *International Joint Conference on Neural Networks (IJCNN)*, 2019.

W. K. Frankenburg, J. Dodds, P. Archer, H. Shapiro, and B. Bresnick. The Denver II: A major revision and restandardization of the Denver Developmental Screening Test. *Pediatrics*, 89(1):91–97, 1992.

M. Gladstone, G. A. Lancaster, E. Umar, M. Nyirenda, E. Kayira, N. R. van den Broek, and R. L. Smyth. The Malawi Developmental Assessment Tool (MDAT): The creation, validation, and reliability of a tool to assess child development in rural African settings. *PLoS Medicine*, 7(5):e1000273, 2010.

R. Griffiths. *The Abilities of Babies.* University of London Press, 1954.

D. P. Kingma and M. Welling. Auto-encoding variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations (ICLR)*, 2014.

F. M. Lord. *Applications of Item Response Theory to Practical Testing Problems.* Lawrence Erlbaum Associates, 1980.

F. M. Lord and M. R. Novick. *Statistical Theories of Mental Test Scores.* Addison-Wesley, 1968.

G. Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests.* Danish Institute for Educational Research, 1960.

R. A. Rigby and D. M. Stasinopoulos. Generalized additive models for location, scale and shape. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 54(3):507–554, 2005.

C. P. Robert and G. Casella. *Monte Carlo Statistical Methods.* Springer, 2nd edition, 2004.

D. L. Robins, K. Casagrande, M. Barton, C.-M. A. Chen, T. Dumont-Mathieu, and D. Fein. Validation of the Modified Checklist for Autism in Toddlers, Revised with Follow-up (M-CHAT-R/F). *Pediatrics*, 133(1):37–45, 2014.

F. Samejima. Graded response model. In W. J. van der Linden and R. K. Hambleton, editors, *Handbook of Modern Item Response Theory*, pages 85–100. Springer, 1997.

J. Squires and D. Bricker. *Ages and Stages Questionnaires, Third Edition (ASQ-3).* Brookes Publishing, 2009.

E. Tsutsumi, R. Kinoshita, and M. Ueno. Deep item response theory as a novel test theory based on deep learning. *Electronics*, 10(9):1020, 2021.

M. Wu, R. L. Davis, B. W. Domingue, C. Piech, and N. Goodman. Variational item response theory: Fast, accurate, and expressive. *Proceedings of the International Conference on Educational Data Mining (EDM)*, 2020.